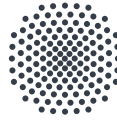




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## **Research Project**

# **Parametric TCAD-Based Investigation of 1200 V Class GaN Power HEMTs on Silicon and Sapphire Substrates**

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## Abstract

Lateral gallium nitride (GaN) high-electron-mobility transistors (HEMTs) grown on silicon substrates are widely adopted for power switching applications due to their low cost and compatibility with established semiconductor manufacturing infrastructure, yet the electrical conductivity of the silicon body imposes two compounding limitations: a hard ceiling on breakdown voltage through a direct vertical punch-through path, and an elevated output capacitance through confinement of the effective dielectric plate separation to the GaN buffer thickness alone. This research project investigates the performance of lateral GaN HEMTs targeting the 1200 V power switching class through two-dimensional technology computer-aided design (TCAD) simulation in Silvaco Victory Device, executed through the nanoHUB.org cloud simulation platform, conducting a parametric sweep across gate-to-drain spacing ( $L_{gd}$ ), GaN buffer thickness, and substrate thickness on both silicon and sapphire substrate platforms, with a dual gate-and-source-connected field plate configuration additionally evaluated for its influence on electric field distribution, output capacitance, and switching figures of merit. No silicon configuration achieves the 1200 V qualification threshold across the entire sweep. The optimal sapphire configuration ( $L_{gd} = 10\text{ }\mu\text{m}$ ,  $0.5\text{ }\mu\text{m}$  buffer,  $500\text{ }\mu\text{m}$  substrate) achieves  $V_{br} = 1814\text{ V}$  at  $R_{ds,on} = 30.86\text{ }\Omega\cdot\text{mm}$ , clearing the threshold with a 51.2% margin, while its silicon counterpart at identical geometry reaches only 754 V at  $31.2\text{ }\Omega\cdot\text{mm}$ , confirming that substrate material alone governs breakdown capability while the conduction path remains unchanged. Even the best performing silicon configuration across the entire sweep ( $L_{gd} = 20\text{ }\mu\text{m}$ ,  $5.0\text{ }\mu\text{m}$  buffer,  $V_{br} = 1043\text{ V}$ ,  $R_{ds,on} = 89.7\text{ }\Omega\cdot\text{mm}$ ) falls 74% below the optimal sapphire breakdown voltage and 190% worse in on-state resistance, despite requiring double the drift region length and a ten-fold thicker buffer. Fifteen of eighteen sapphire configurations satisfy the 1200 V threshold, with breakdown voltages reaching up to 2958 V and  $C_{o(er)}$  reduced by up to 31.9% relative to silicon at the 1200 V evaluation bias, while the optimal configuration additionally offers manufacturing cost advantages through a thin epitaxial buffer that reduces Metal-Organic Chemical Vapor Deposition (MOCVD) reactor time and a substrate thickness matching the standard as-grown commercial wafer dimension, eliminating post-growth thinning entirely. The integration of a dual gate-and-source-connected field plate on the sapphire platform suppresses the gate-edge electric field peak from above  $5.0 \times 10^6\text{ V/cm}$  to approximately  $1.0 \times 10^6\text{ V/cm}$ , reduces  $C_{o(er)}$  and stored output energy  $E_{o(er)}$  above 400 V, and is expected to yield higher physical breakdown voltage in a fabricated device, establishing a three-tier performance hierarchy in which silicon fails unconditionally, sapphire meets and exceeds the voltage target with improved switching efficiency, and the dual field plate further improves lateral field management and reduces switching energy losses.

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# 1 Introduction

## 1.1 Motivation

The rapid electrification of transportation and renewable energy infrastructure is fundamentally reshaping the requirements placed on power semiconductor technology. Electric vehicles are migrating from 400 V to 800 V traction architectures to enable faster charging and higher power density, while grid-scale photovoltaic and battery storage installations are converging on 1500 V bus voltages to minimize transmission losses and balance-of-system cost [1], [2]. These shifts place semiconductor switches capable of reliably blocking 1200 V and above at the center of every power converter that moves energy between the grid, storage, and the electric drivetrain.

The material that has long promised to meet this challenge most efficiently is GaN. The theoretical performance ceiling of a semiconductor switch is quantified by the Baliga figure of merit, which scales with the cube of the critical electric field [3]. GaN possesses a critical field of approximately 3.3 MV/cm, roughly ten times that of silicon, placing it among the most capable wide-bandgap materials known for power switching applications [3], [4]. The physical origin of this capability lies in the unique properties of the wurtzite crystal structure: spontaneous and piezoelectric polarization discontinuities at the AlGaIn/GaN heterointerface induce a high-density, high-mobility two-dimensional electron gas (2DEG) without intentional doping, enabling sheet carrier densities on the order of  $10^{13} \text{ cm}^{-2}$  [5]–[7]. This combination of high carrier density, high mobility, and wide bandgap gives GaN HEMTs a fundamental advantage over both silicon and silicon carbide (SiC) in switching applications [1], [4]. To translate this material potential into reliable high-voltage devices, the lateral electric field in the drift region must be carefully managed through electrostatic shaping structures, a challenge whose analytical foundation was laid by Karmalkar and Mishra [8].

At the 600 V class, commercial GaN-on-Si devices already demonstrate this promise in practice, outperforming competing silicon technology on on-state resistance, switching speed, and passive component size while retaining the cost and infrastructure advantages of large-diameter silicon wafer manufacturing [1], [9]. SiC metal-oxide-semiconductor field-effect transistors (MOSFETs) currently dominate the 1200 V class, handling the electrical stress competently but at a manufacturing cost that remains substantially higher than silicon-based alternatives [1], [2]. Given how decisively GaN has outperformed silicon at 600 V, and given the scale of the economic opportunity in the 1200 V market

driven by electrification, it is both a scientifically significant and industrially consequential question whether the same device physics can be scaled to meet the demands of next-generation carbon-free energy systems [2], [10].

## 1.2 Literature Review and State of the Art

Silicon remains the dominant substrate for GaN manufacturing because of its availability in large wafer diameters, its established processing infrastructure, and its comparatively low cost [1], [11]. However, the heteroepitaxial growth of GaN on silicon is fundamentally constrained by severe material incompatibilities, specifically a lattice mismatch of approximately 17% and a thermal expansion coefficient mismatch of roughly 54% [10], [11]. This massive disparity introduces immense tensile stress into the epitaxial stack during cooling, restricting the practical thickness of the GaN buffer layer to approximately 5  $\mu\text{m}$  before catastrophic cracking or severe wafer bow occurs [11], [12].

Because the buffer layer is responsible for vertical voltage isolation, this physical thickness ceiling translates directly into a hard blocking voltage ceiling [11], [12]. Furthermore, the conductive silicon body beneath the buffer acts as an active electron source. At high reverse biases, electrons are injected from the silicon substrate into the buffer layer via trap-assisted tunneling and Poole-Frenkel emission, creating a parallel vertical leakage path that causes premature vertical punch-through well before the 1200 V threshold is reached [11], [13], [14].

To suppress this vertical leakage, intentional carbon doping is widely employed during epitaxial growth to introduce deep-level acceptor traps, typically located around 0.9 eV above the valence band, which compensate for background donor impurities and raise buffer resistivity [14]–[16]. While these deep acceptors delay vertical punch-through, they trap injected electrons under high off-state stress and create a negative space charge at the interface that depletes the active 2DEG from the backside. This trap-induced back-gating effect drastically inflates the dynamic on-state resistance ( $R_{ds,on}$ ) during high-frequency switching operations [15], [16]. Beyond these conduction penalties, the conductive silicon substrate acts as a parallel capacitor plate separated from the channel by only the thin buffer layer, which severely inflates the parasitic output capacitance ( $C_{oss}$ ) and introduces substantial switching energy losses [17].

Even if the vertical constraint were resolved entirely, the device would remain limited by its lateral electrostatics. In a standard lateral HEMT geometry, the off-state electric field is not distributed uniformly across the gate-to-drain drift region; instead, it concentrates sharply at the drain-side edge of the gate electrode. Under high reverse bias, this extreme field crowding triggers localized impact ionization or progressively degrades the surface passivation dielectric, causing premature lateral breakdown long before the theoretical limit of the drift region material is reached [1], [8], [22]. This lateral field crowding is

therefore an independent and equally fundamental bottleneck that must be addressed in parallel with the substrate limitation.

## 1.3 Problem Statement

The obstacle preventing GaN from repeating its 600 V success at higher voltages is not a limitation of the material itself [1], [10]. Instead, it arises from two distinct and independent physical bottlenecks introduced by the device architecture and substrate choice. When scaling lateral HEMTs to the 1200 V regime, both mechanisms must be overcome simultaneously:

1. **Vertical Failure:** The epitaxial thickness ceiling imposed by the GaN-on-Si material system limits the vertical blocking strength of the buffer layer. The thin buffer, combined with the highly conductive silicon body acting as an active electron source beneath it, causes substrate-induced leakage and vertical punch-through well below 1200 V [11], [13], [14]. Carbon-doped buffer strategies suppress this leakage but introduce dynamic  $R_{ds,on}$  degradation and inflate  $C_{oss}$  [15]–[17].
2. **Lateral Failure:** Even on an ideal insulating substrate, the lateral electric field in a standard HEMT peaks sharply at the drain-side gate edge under reverse bias. This localized field spike triggers avalanche breakdown through impact ionization or damages the passivation dielectric, causing catastrophic lateral failure far below the theoretical capacity of the drift region [1], [8], [22].

Resolving either bottleneck in isolation does not yield a viable 1200 V device; both must be addressed jointly. This dual requirement defines the central engineering challenge that this thesis investigates.

## 1.4 Research Questions

The primary objective of this research project is to assess, through technology computer-aided design (TCAD) simulations, the extent to which lateral GaN HEMT technology can be pushed beyond the 1200 V barrier by jointly varying the substrate, device geometry, and field plate architecture. This study seeks to answer the following specific research questions:

1. **Addressing Vertical Failure:** How does replacing the conventional conductive silicon substrate with an electrically insulating sapphire substrate eliminate the vertical punch-through limit [11], [13], and how does removing the conductive body alter the parasitic equivalent output capacitance ( $C_{o(er)}$ ) [17]?

2. **Addressing Lateral Failure (Geometry):** What is the quantitative relationship between extending the gate-to-drain spacing ( $L_{gd}$ ) and the resulting improvement in blocking voltage ( $V_{br}$ ), and what is the accompanying penalty on conduction losses ( $R_{ds,on}$ ) [3], [8], [22]?
3. **Addressing Lateral Failure (Electrostatics):** To what extent do multi-stepped field plate configurations successfully redistribute the peak electric field at the gate edge [8], [26], [27], and how do these structures modify the dynamic switching profile through their effect on terminal capacitances and charge ( $C_{oss}$ ,  $Q_{gd}$ ) [25], [29]?

By answering these questions within a shared TCAD simulation framework, this work aims to identify the structural configuration that offers the most favourable balance among blocking voltage, conduction losses, and switching efficiency, and to establish which combination of substrate, geometry, and field plate architecture is best suited for 1200 V lateral GaN HEMT operation.

## 2 Strategies for 1200 V GaN HEMT Operation

The two failure mechanisms identified in Chapter 1 (substrate-induced vertical Failure and Lateral Failure due to Electrical Field peaks) are each the subject of a distinct and well-established body of engineering literature. This chapter reviews the physical principles and reported results behind three complementary strategies that have been proposed to overcome them: substrate engineering for vertical isolation, gate-to-drain geometry scaling for lateral field distribution, and field plate integration for electrostatic field shaping. Each strategy is reviewed independently before the final section identifies the gap in existing knowledge that the present work addresses.

### 2.1 Substrate Engineering for Vertical Isolation

The most direct route to eliminating the vertical punch-through limitation is to remove the conductive body beneath the buffer entirely. Replacing silicon with an electrically insulating substrate removes the parallel leakage path, allows the substrate itself to contribute to vertical voltage isolation, and substantially relaxes the epitaxial thickness constraint that prevents GaN-on-Si from reaching 1200 V [11], [20].

Sapphire is the most established electrically insulating alternative for power-class GaN HEMTs. Its intrinsic electrical resistivity is sufficient to suppress vertical leakage to negligible levels across the full voltage range of interest, and it is available at substantially lower cost than silicon carbide [20], [21]. Gupta et al. demonstrated 1200 V blocking on a GaN-on-sapphire platform, confirming that the substrate is physically capable of sustaining this voltage class [20]. Cheng et al. further demonstrated that introducing carbon doping into the GaN buffer grown on sapphire provides an additional means of suppressing residual leakage, with the carbon atoms acting as deep-level compensating centers that raise buffer resistivity without introducing the back-gating penalty associated with the same technique on silicon since the conductive backplane that drives back-gating on silicon is absent [15], [21]. Beyond these electrical advantages, removing the conductive silicon body also eliminates the parasitic capacitor plate effect that inflates  $C_{oss}$  on silicon substrates, with a direct reduction in switching energy losses [17], [20].

Engineered and composite substrates represent a more recent research direction aimed at matching the thermal expansion coefficient of GaN more closely than sapphire achieves, enabling thicker and higher-quality epitaxial buffers on larger wafer diameters that sapphire has historically not made straightforward [10], [11]. Across all insulating substrate options, the trade-off between thermal conductivity, lattice matching, substrate cost, and

wafer scalability is consistent: no single alternative dominates on all axes simultaneously [11].

## 2.2 Gate-to-Drain Spacing and the $V_{br}$ – $R_{ds,on}$ Trade-Off

Once the vertical constraint is resolved through substrate engineering, the dominant limitation on blocking voltage becomes the distribution of the lateral electric field in the gate-to-drain drift region. The gate-to-drain length ( $L_{gd}$ ) is the most direct geometric parameter for controlling this distribution. A longer drift region spreads the lateral electric field over a greater distance, reduces the peak field intensity at the gate edge, and directly raises the avalanche breakdown voltage ( $V_{br}$ ) [8], [22], [23]. Saito et al. were among the first to systematically demonstrate this relationship in high-voltage AlGaIn/GaN HEMTs, reporting breakdown voltages scaling broadly in proportion to drift region length across devices spanning the 1 kV class [22].

The trade-off is equally fundamental. The same drift region that sustains the off-state voltage also carries the on-state current, and its sheet resistance contributes directly to the total  $R_{ds,on}$ . Increasing  $L_{gd}$  therefore raises conduction losses in proportion to the added drift length [22], [24]. This relationship is captured by the Baliga figure of merit, which defines the theoretical upper bound on the  $V_{br}^2/R_{ds,on}$  product achievable for a given semiconductor material and device architecture [3], [24]. Every additional micron of drift region improves  $V_{br}$  but moves the operating point along this trade-off curve, and no geometry can exceed the bound it sets.

Critically, the slope of the  $V_{br}$ – $L_{gd}$  relationship is not fixed by geometry alone; it depends strongly on the field plate configuration applied over the drift region. A field plate reshapes the electric field distribution and allows a given  $L_{gd}$  to sustain higher voltage than the same geometry without one, meaning that  $L_{gd}$  and field plate architecture must be characterized jointly rather than in isolation [8], [24], [25].

## 2.3 Field Plate Engineering for Electric Field Shaping

Field plates are the primary electrostatic engineering tool for suppressing the gate-edge field peak that initiates premature lateral breakdown. A field plate is a metal extension placed over part of the gate-to-drain access region, separated from the semiconductor surface by a dielectric, and connected to the gate, source, or drain terminal. By extending the region over which the potential transitions from the gate to the drain, the plate redistributes the electric field laterally and reduces its peak value at the gate edge.

### 2.3.1 Single Gate-Connected Field Plate

The foundational analytical treatment by Karmalkar and Mishra established that a single, appropriately dimensioned gate-connected field plate can raise the breakdown voltage of a lateral HEMT substantially relative to an otherwise identical device without one [8]. The mechanism is the introduction of a second electric field peak at the plate tip: by controlling the plate length and dielectric thickness, the gate-edge peak and the plate-tip peak can be equalized and kept simultaneously below the critical field, roughly doubling the usable voltage range [8]. Wu et al. provided the first experimental confirmation of large-signal performance gains from this approach, demonstrating 30 W/mm output power density in GaN HEMTs with an optimized gate field plate [26].

The same analysis identified the intrinsic saturation of the single-plate approach: once the two peaks are equalized, further extending the plate does not continue to raise the breakdown voltage, because the plate-tip peak does not decrease with additional plate extension. A performance ceiling is therefore reached with a single field plate, and overcoming it requires a multi-plate architecture [8].

### 2.3.2 Multi-Step Field Plate Configurations

Distributing the field-shaping task across multiple plates overcomes the single-plate saturation by introducing additional transition zones across the drift region. In a double field-plate configuration, a gate-connected plate manages the field immediately beyond the gate while a source-connected plate extends further into the drift region, suppressing both the primary gate-edge peak and the secondary plate-tip peak and pushing the overall field profile toward uniformity [23], [24], [27]. Wu et al. demonstrated 40 W/mm output power density with a double field-plate architecture, confirming the performance gain over the single-plate approach [27]. At the 1200 V power switching level, this configuration has been shown to enable reliable blocking with simultaneously suppressed dynamic  $R_{ds,on}$  degradation [25], [28], which is a critical requirement for high-frequency power converters where trap-induced resistance increase under off-state stress directly degrades efficiency.

Each additional plate, however, introduces parasitic capacitance between its terminal and the drain, increasing gate charge ( $Q_{gd}$ ) and equivalent output capacitance ( $C_{o(er)}$ ) and penalizing switching speed [25], [29]. Achieving 1200 V operation with acceptable switching performance therefore requires optimizing field plate number and geometry jointly with  $L_{gd}$ , rather than maximizing the number of plates without regard to dynamic performance.

### 2.3.3 Slanted Field Plates

A geometrically distinct variant (the slanted or inclined field plate, in which the plate metallization rises at an angle toward the drain rather than extending horizontally) has been demonstrated to achieve comparable breakdown voltage improvements with lower incremental parasitic capacitance than flat multi-plate structures [30]. This approach is particularly relevant in applications where switching frequency is as important as blocking capability, since the reduced  $C_{o(er)}$  directly translates into lower switching energy loss per cycle.

## 2.4 Research Objective and Simulation Approach

The body of literature reviewed above establishes that substrate engineering, gate-to-drain geometry scaling, and field plate architecture are each individually effective strategies for extending the operating voltage of lateral GaN HEMTs. However, these variables have predominantly been studied in isolation. Individual experimental studies typically fix the substrate and field plate configuration and vary only  $L_{gd}$ , or fix the geometry and compare field plate configurations on a single substrate [23]–[25]. The interaction effects between these variables (specifically how the substrate’s removal of the vertical constraint changes the sensitivity of  $V_{br}$  to  $L_{gd}$ , or how the optimal field plate configuration shifts when the drift region length changes) have not been systematically characterized under a common set of conditions. This gap in joint parameter evaluation is the methodological limitation that the present work directly addresses.

Characterizing these interactions empirically through iterative physical fabrication is prohibitively time-consuming and expensive. A computational approach using technology computer-aided design (TCAD) is therefore strictly necessary to isolate each variable, control all others, and evaluate the full combination space within a consistent physical framework [18], [19]. The research framework adopted in this work is illustrated in Figure 2.1.

The parametric sweep is structured around three independent design variables, each varied across a defined range while all other parameters are held constant:

- **Substrate Technology:** Conventional conductive silicon serves as the baseline and is benchmarked against an electrically insulating sapphire substrate to isolate the impact of the vertical conductive body on breakdown voltage and terminal capacitance. The vertical epitaxial stack is further explored by sweeping the buffer thickness (1  $\mu\text{m}$  and 5  $\mu\text{m}$ ) and the substrate thickness (5  $\mu\text{m}$ , 50  $\mu\text{m}$ , and 500  $\mu\text{m}$ ) to quantify how these dimensions influence vertical blocking and  $C_{oss}$ .
- **Gate-to-Drain Spacing ( $L_{gd}$ ):** The lateral drift region is varied across three values (10  $\mu\text{m}$ , 15  $\mu\text{m}$ , and 20  $\mu\text{m}$ ) to characterize the  $V_{br}$ – $R_{ds,on}$  trade-off and the Baliga figure of merit boundary for this device class.

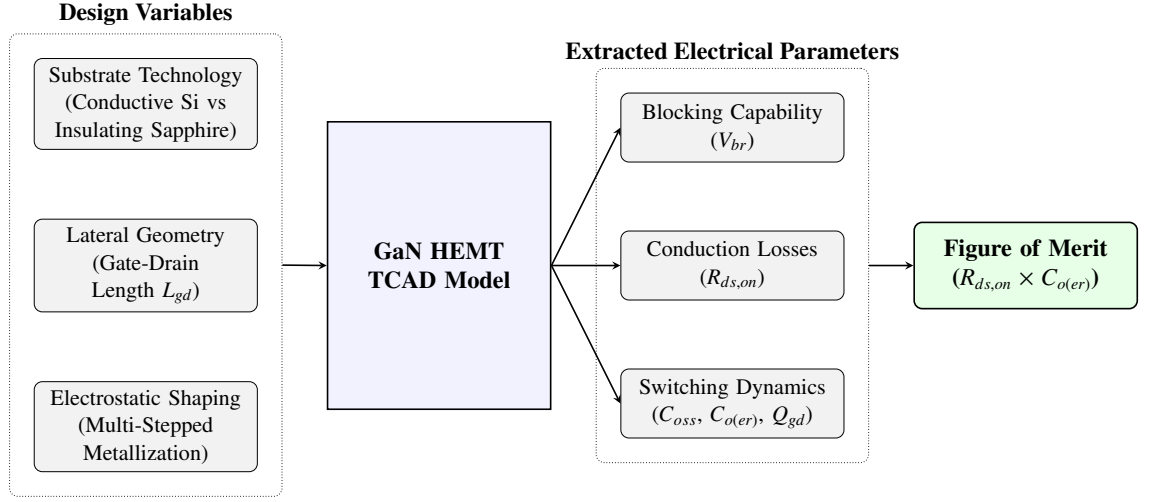


Figure 2.1: Research framework mapping the three design variables to the extracted electrical figures of merit evaluated in this work.

- **Field Plate Architecture:** The metallization configuration is varied across three designs (no field plate, a single gate-connected field plate, and a combined gate-connected and source-connected double field plate) to study how each configuration redistributes the peak electric field and modifies the dynamic switching profile.

Rather than optimizing for a single figure of merit, every combination in this sweep is characterized across a comprehensive and identical set of output metrics: output characteristics ( $I_d$ – $V_{ds}$ ), transfer characteristics ( $I_d$ – $V_{gs}$ ), capacitance-voltage behavior, breakdown voltage,  $R_{ds,on}$ , and the full suite of charge and energy-related capacitance metrics ( $C_{o(er)}$ ,  $C_{o(tr)}$ ,  $Q_{gd}$ ). This ensures a direct and rigorous like-for-like comparison across every proposed solution. The baseline device cross-section that forms the starting point for this sweep is illustrated in Figure 2.2.

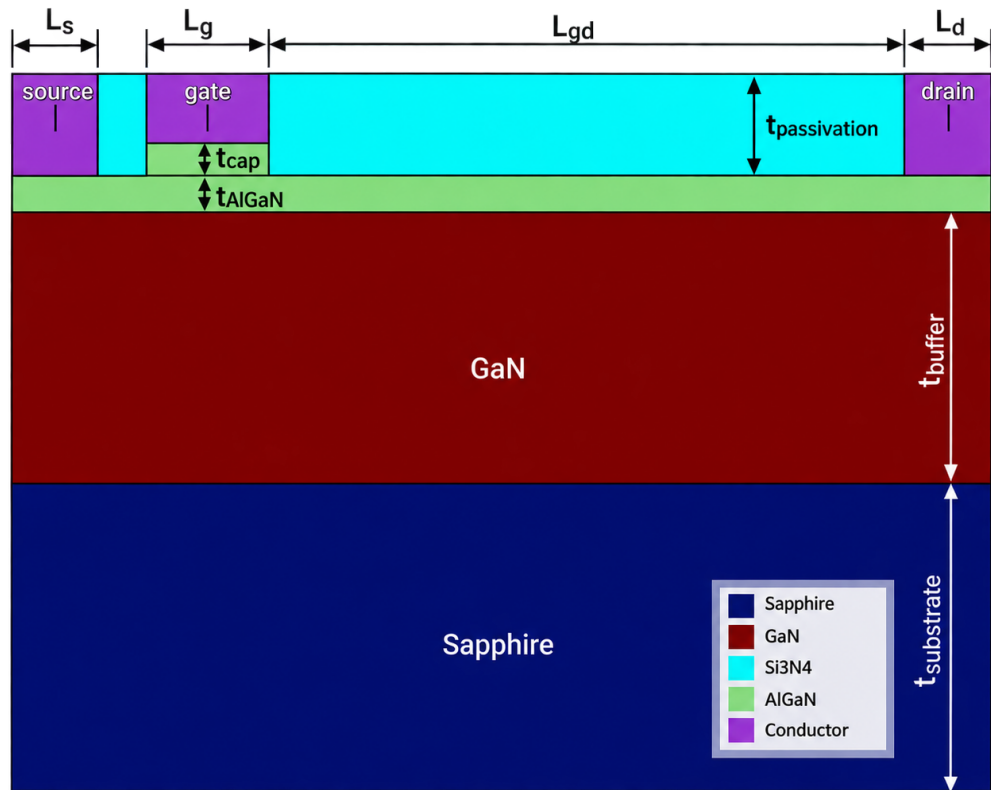


Figure 2.2: 2D cross-section of the simulated GaN HEMT architecture outlining the design and primary geometric variables of the parametric sweep.

### 3 Simulation Framework and Methodology

The device model is implemented in Silvaco Victory Device, a two-dimensional technology computer-aided design (TCAD) simulator capable of self-consistently solving the drift-diffusion equations alongside the polarization, trap, and impact-ionization physics required for wide-bandgap heterostructure devices. To systematically evaluate the high-voltage scaling of these GaN HEMTs across multiple geometric and material variations, a highly automated, end-to-end simulation and post-processing pipeline was engineered. The complete architecture of this workflow is illustrated in Figure 3.1.

As depicted in the workflow diagram, the framework is divided into three distinct execution phases to ensure high throughput and strict repeatability across the design of experiments (DoE) sweep:

1. **Deck Generation (Local PC):** The pipeline is initialized locally using a custom Python script. This generator merges a standardized TCAD template file with an array of predefined physical parameters, such as substrate material, gate-to-drain spacing ( $L_{gd}$ ), and buffer thickness. By programmatically generating the simulation decks, human error is eliminated, ensuring that every device variant is initialized with identical meshing strategies and baseline physics.
2. **TCAD Simulation (nanoHUB Virtual PC):** The generated decks are subsequently executed on the nanoHUB computational cluster. nanoHUB serves as a cloud-based platform that provides free, remote access to commercial-grade simulation software, bypassing expensive licensing fees. Furthermore, utilizing these high-performance computing resources enables the highly efficient execution of the computationally intensive numerical analysis required for this DoE. Within this environment, the simulation sequence strictly follows a defined internal hierarchy: dynamic mesh generation, structural and epitaxial definition, electrode assignment, and finally, the activation of the physical models. Once the device is virtually fabricated, the solver executes four parallel characterization routines on every variant: output characteristics ( $I_d-V_{ds}$ ), transfer characteristics ( $I_d-V_{gs}$ ), small-signal alternating current (AC) capacitance, and high-voltage breakdown sweeps.
3. **Post-Processing and Output Generation (Local PC):** Upon completion, the raw simulation logs are downloaded back to the local environment. A secondary suite of custom Python scripts parses these dense log files, filtering out numerical artifacts and applying rigorous mathematical integration. This final stage automatically extracts the primary figures of merit required for power electronic evaluation, specifically the specific on-state resistance ( $R_{ds,on}$ ), the breakdown voltage ( $V_{br}$ ),

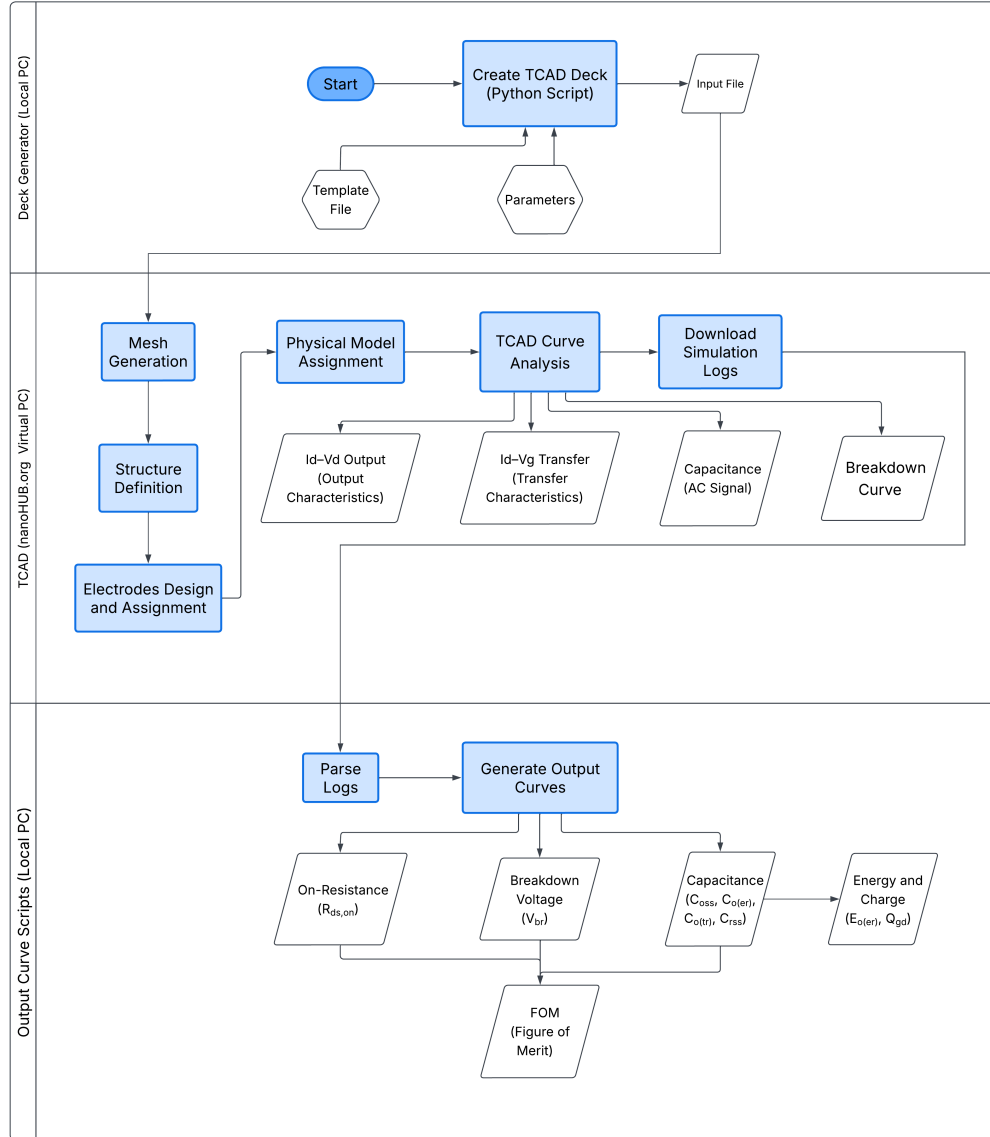


Figure 3.1: Overview of the automated TCAD simulation and post-processing workflow.

the terminal and equivalent energy-related capacitances ( $C_{oss}$ ,  $C_{o(er)}$ ,  $C_{o(tr)}$ ,  $C_{rss}$ ), and the dynamic charge and energy losses ( $Q_{gd}$ ,  $E_{o(er)}$ ).

By utilizing this parameterized pipeline, substrate material, gate-to-drain spacing, buffer thickness, and field-plate configurations can each be varied independently while every other aspect of the extraction procedure remains fixed. This ensures that differences observed in the final output curves can be conclusively attributed to the swept design parameter rather than to inconsistencies in the simulation setup itself.

### 3.1 Device Geometry

To accurately model the physical domains and boundaries of the device, two-dimensional cross-sectional structures were generated in Silvaco Victory Device. The specific geometries and epitaxial stacks for the primary device variants are illustrated through a progression of macro- and microscopic schematics.

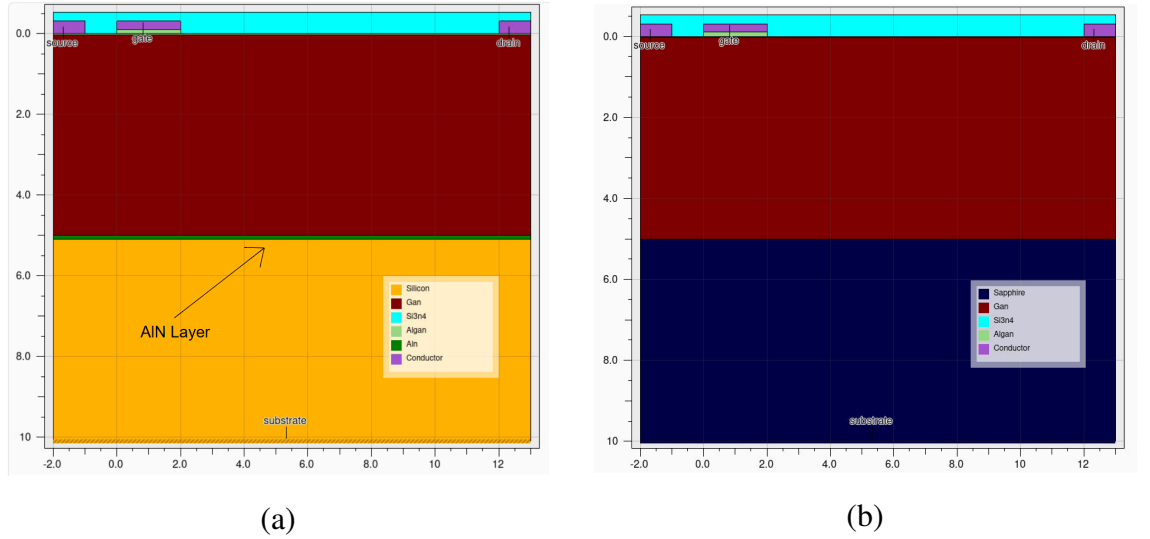


Figure 3.2: Cross-sectional schematics of the (a) Silicon Substrate and (b) Sapphire Substrate architectures.

The baseline architectures comparing the conductive silicon substrate and the insulating sapphire substrate variant are presented side-by-side in Figure 3.2. The silicon baseline structure is built upon a conductive silicon substrate featuring an aluminum nitride (AlN) nucleation layer between the substrate and the GaN buffer to manage lattice mismatch and prevent melt-back etching. Above this, the thick GaN buffer and AlGaIn barrier form the active region, encapsulated by a silicon nitride ( $\text{Si}_3\text{N}_4$ ) passivation layer. To isolate the vertical punch-through failure mechanism, the conductive silicon is replaced with an electrically insulating sapphire substrate. The geometric dimensions of the active drift

region ( $L_{gd}$ ), buffer, and passivation remain completely identical to ensure a rigorous like-for-like comparison, effectively eliminating the underlying conductive body.

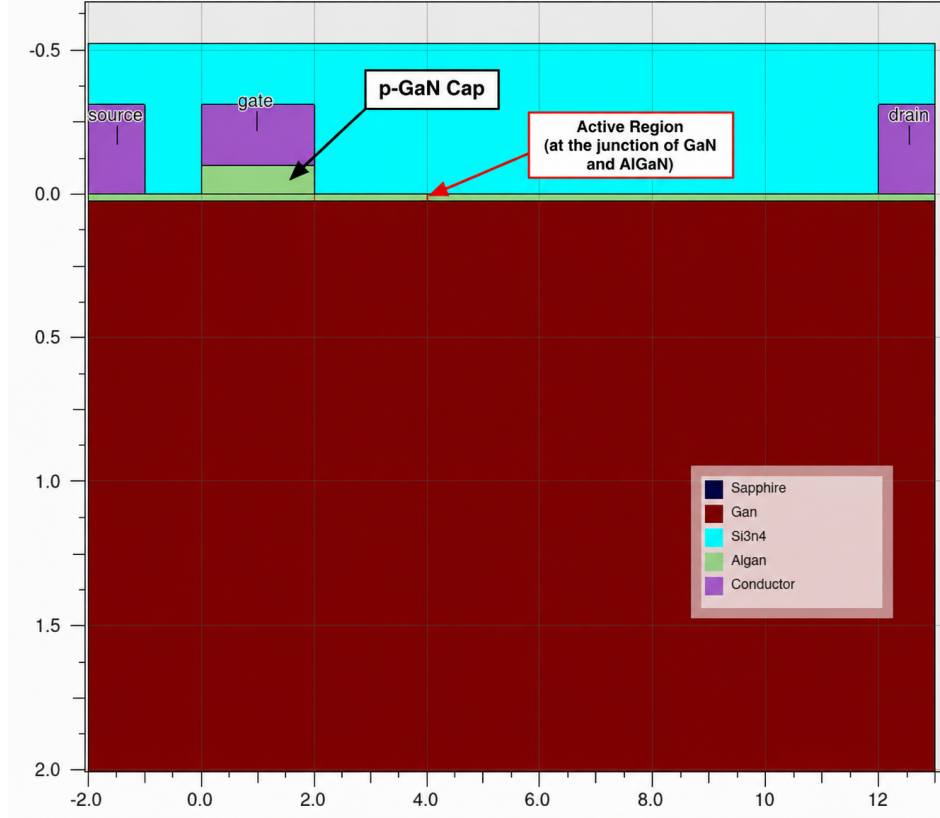


Figure 3.3: Cross-section of the active channel and p-GaN cap gate structure.

While macroscopic cross-sections illustrate the deep vertical isolation domains, the high-field carrier dynamics are governed by the immediate vicinity of the active channel. A detailed close-up of this critical active region is presented in Figure 3.3. Here, the heterojunction is formed by a 25 nm AlGaN barrier layer grown over the GaN buffer. To achieve normally-off enhancement-mode operation, a 100 nm p-doped AlGaN cap layer is selectively grown and recessed directly beneath the gate electrode. This recessed p-gate physically lifts the conduction band above the Fermi level at zero bias, depleting the underlying 2DEG. The entire access region between the source, gate, and drain contacts is passivated by a thick  $\text{Si}_3\text{N}_4$  layer to mitigate surface-state-induced current collapse.

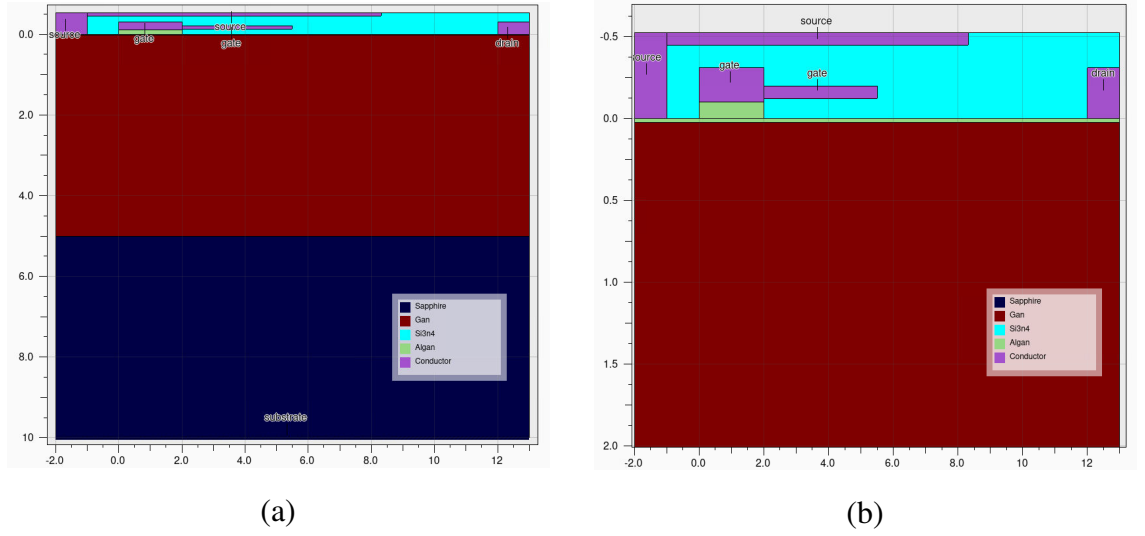


Figure 3.4: Macro and micro geometries of the (a) GaN HEMT with integrated gate and source field-plate (FP) Architecture and (b) Detailed Zoomed-In FP Geometry.

To further modulate the electrostatic profile of the active region, field-plate architectures were introduced into the simulation. The macro-level integration alongside the precise multi-stepped geometry is compared side-by-side in Figure 3.4. A gate field plate and a source field plate are integrated into the device by stepping over the dielectric layers. The gate metal is extended laterally over the primary Si<sub>3</sub>N<sub>4</sub> passivation toward the drain, while the source metal is subsequently routed over the gate field plate, separated by a secondary inter-layer dielectric, extending further into the drift region to reshape the electric field distribution during high-voltage off-state conditions.

## 3.2 Meshing Strategy

The accuracy of the drift-diffusion and physical models utilized in this study depends critically on the spatial discretization of the device structure. A highly optimized, non-uniform finite element mesh was developed to balance numerical convergence, computational efficiency, and physical precision. The mesh density is dynamically scaled across the device layers, transitioning from a coarse grid in the deep substrate to an ultra-dense grid in the active channel.

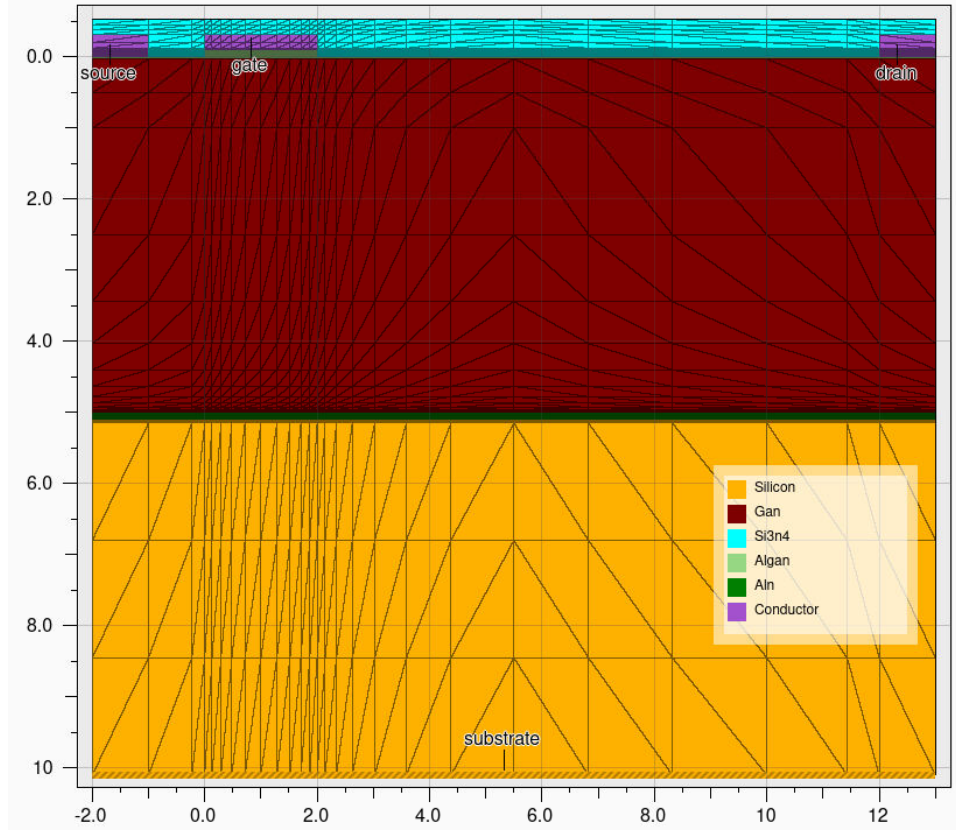


Figure 3.5: Coarse mesh applied across the deep substrate region.

Figure 3.5 illustrates the macroscopic meshing approach across the deep substrate region. In the deep silicon substrate, the spatial variation of the electric potential and carrier concentration is minimal. Because this region acts essentially as a passive dielectric block where the electric potential drops linearly, a highly coarse mesh is applied. This prevents matrix solver memory overflow and significantly reduces computational overhead without compromising the accuracy of the vertical breakdown characteristics.

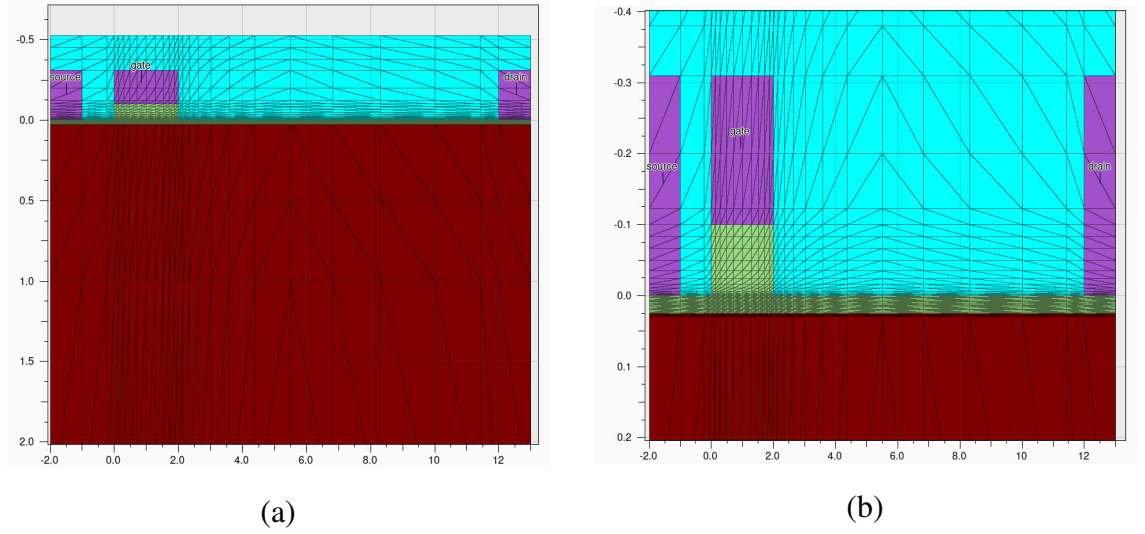


Figure 3.6: Mesh density transition from (a) Dense Mesh for Buffer to (b) Ultra-Dense Mesh for Active Channel and Electrodes.

Moving upward into the epitaxial stack and active heterojunction, the transitional buffer mesh and ultra-dense active channel mesh are presented side-by-side in Figure 3.6. The buffer mesh is explicitly graded, becoming progressively finer toward the top surface. This moderate, graded density is mathematically necessary to accurately resolve the vertical electric field distribution, track vertical leakage, and model the deep acceptor space-charge regions.

Conversely, ultra-fine vertical mesh spacing on the order of nanometers is enforced at the AlGa<sub>N</sub>/Ga<sub>N</sub> interface. The carrier concentration drops from peak density to near-zero over a vertical distance of just a few nanometers, and this nanometer-scale spatial resolution is strictly required to prevent the Poisson equation solver from failing to converge across the sharp quantum well gradient. Furthermore, a heavily concentrated lateral grid density is applied around the recessed p-gate and field-plate edges. If the mesh is too coarse in these terminal regions, the finite element solver artificially smooths out the peak electric field, which drastically overestimates the breakdown voltage. A dense lateral grid is therefore strictly required to capture the true, sharp localized field crowding that triggers avalanche breakdown.

### 3.3 Polarization and Two-Dimensional Electron Gas (2DEG) Formation

The fundamental advantage of the AlGa<sub>N</sub>/Ga<sub>N</sub> heterostructure is the formation of a high-density, high-mobility two-dimensional electron gas (2DEG) at the AlGa<sub>N</sub>/Ga<sub>N</sub> interface

without intentional channel doping. The physical origin of the 2DEG is the strong polarization discontinuity at the heterointerface, which arises from two additive contributions: spontaneous and piezoelectric polarization.

GaN and AlGa<sub>N</sub> crystallize in the wurtzite structure, which lacks inversion symmetry along the  $c$ -axis. This asymmetry gives rise to a built-in electric dipole per unit volume in the unstrained crystal, known as spontaneous polarization ( $P_{SP}$ ). The magnitude of spontaneous polarization differs between AlGa<sub>N</sub> and GaN. In addition, when an AlGa<sub>N</sub> barrier is grown pseudomorphically on a relaxed GaN layer, the lattice mismatch produces in-plane biaxial compressive strain in the AlGa<sub>N</sub> layer. This strain generates an additional piezoelectric polarization ( $P_{PE}$ ). For the conventional Ga-polar AlGa<sub>N</sub>/GaN heterostructure, the piezoelectric contribution is oriented in the same direction as the spontaneous polarization and therefore increases the total polarization.

The total polarization can therefore be expressed as

$$P_{\text{tot}} = P_{SP} + P_{PE}. \quad (3.1)$$

The discontinuity in total polarization across the AlGa<sub>N</sub>/GaN interface produces a fixed polarization sheet charge. For a conventional Ga-polar AlGa<sub>N</sub>/GaN heterostructure, this results in positive polarization charge at the AlGa<sub>N</sub>/GaN interface and compensating negative charge toward the AlGa<sub>N</sub> surface. The associated electric field produces downward bending of the conduction-band edge toward the interface, lowering the conduction-band energy on the GaN side and forming a triangular quantum well.

Electrons can then accumulate in this low-energy region. In the surface-state model proposed for nominally undoped AlGa<sub>N</sub>/GaN heterostructures, electrons supplied by ionized surface donor states transfer toward the AlGa<sub>N</sub>/GaN interface and accumulate in the triangular quantum well. The electrons are confined within a narrow region near the interface, with their motion quantized in the direction perpendicular to the interface while remaining relatively free in the plane of the interface. This spatial confinement gives rise to a two-dimensional electron gas (2DEG). The relationship between the polarization-induced charge distribution, band bending, triangular quantum well, and electron accumulation is illustrated in Figure 3.7 [5], [7].

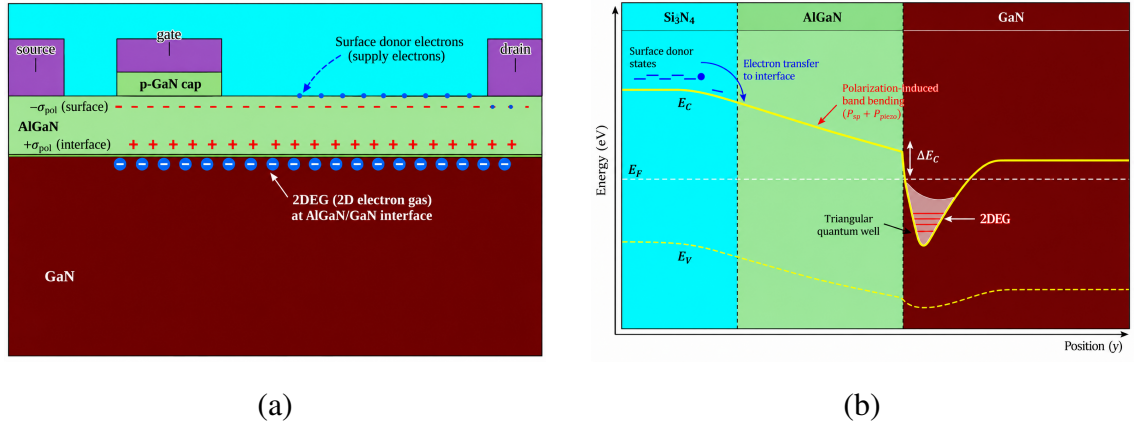


Figure 3.7: Schematic illustration of 2DEG formation in an AlGaIn/GaN heterostructure, showing (a) Charge Distribution and Surface States and (b) polarization-induced band bending, the triangular quantum well, and electron transfer from surface donor states, adapted from [5], [7].

Because the 2DEG is formed without intentional donor doping in the channel, ionized-impurity scattering associated with intentional channel doping is substantially reduced. Consequently, AlGaIn/GaN heterostructures can exhibit high room-temperature electron mobilities together with sheet carrier densities typically of the order of  $10^{13} \text{ cm}^{-2}$  [4], [6].

In the simulation deck, the spontaneous and piezoelectric polarization models are activated using:

```
MODELS POLARIZATION psp.scale=0.6 piezo.scale=0.6 calc.strain
```

The `calc.strain` option enables the calculation of strain arising from lattice mismatch between the epitaxial layers. Rather than assuming a fixed strain state, the simulator accounts for the lattice mismatch between the constituent materials when evaluating the piezoelectric polarization.

The spontaneous and piezoelectric polarization contributions are each scaled to 60% of their theoretical values in the simulation. This scaling is used as an effective calibration to account phenomenologically for the reduction of the ideal polarization-induced charge in the simulated heterostructure due to non-ideal effects such as incomplete strain relaxation and charge compensation. The scaling factor was selected to obtain a simulated 2DEG sheet density consistent with experimentally reported values for comparable  $\text{Al}_{0.15}\text{Ga}_{0.85}\text{N}/\text{GaN}$  heterostructures.

The resulting conduction-band profile, valence-band profile, total charge distribution, and polarization charge distribution are exported for post-processing using:

OUTPUT con.band val.band charge polar.charge

These outputs are used to examine the relationship between polarization charge, energy-band bending, and electron accumulation at the AlGa<sub>N</sub>/Ga<sub>N</sub> interface.

The lateral transport of 2DEG electrons in the on-state is governed by the gansat.n albrct.n mobility model, which combines the Albrecht formulation for carrier mobility with Ga<sub>N</sub>-specific velocity-saturation behaviour. Under an applied drain bias, electrons confined in the 2DEG move laterally along the AlGa<sub>N</sub>/Ga<sub>N</sub> interface from the source toward the drain, providing the principal conduction path of the HEMT.

In the p-Ga<sub>N</sub> gate HEMT, the gate electrode is formed above a p-Ga<sub>N</sub> cap. The p-Ga<sub>N</sub> gate structure enables normally-off operation by controlling the electron population in the underlying 2DEG. In the off-state, the gate potential depletes the 2DEG beneath the gate region, thereby interrupting the conduction path between the source and drain. The p-Ga<sub>N</sub> gate cap in the present simulation corresponds to region=5 and is uniformly doped with a p-type concentration of  $4 \times 10^{18} \text{ cm}^{-3}$ .

When the device is turned on by applying an appropriate positive gate bias, depletion of the 2DEG beneath the gate is reduced and a continuous conducting channel is established between the source and drain. The resulting current is therefore primarily carried by the high-density 2DEG along the AlGa<sub>N</sub>/Ga<sub>N</sub> interface.

## 3.4 Physical Models

### 3.4.1 Recombination: SRH and Auger

Carrier recombination is modeled through Shockley-Read-Hall (SRH) and Auger processes (srh auger). SRH recombination captures trap-assisted recombination through deep-level defects in the Ga<sub>N</sub> buffer and is particularly relevant to the acceptor trap defined in the deck (DOPING trap acceptor e.level=1.0 concentration=4e18), which represents the carbon-doped buffer layer. These deep acceptor traps raise buffer resistivity and suppress vertical leakage, which is critical for blocking high drain voltages. Auger recombination, in which the energy of a recombining electron-hole pair is transferred to a third carrier, becomes relevant at the high carrier densities encountered near breakdown.

### 3.4.2 Electron and Hole Mobility

Electron transport in the Ga<sub>N</sub> channel is governed by the Albrecht phonon-scattering model combined with Ga<sub>N</sub>-specific velocity saturation (gansat.n albrct.n). This model captures the field-dependent reduction in electron drift velocity at high lateral

fields in the drift region, which directly determines the on-state resistance and the current saturation behavior. Hole transport in the p-GaN gate cap is modeled separately via `albrct.p`. In the breakdown remesh, the Albrecht coefficients are slightly adjusted (`bn.albrct=7e-04` vs. `5e-04`) to improve numerical convergence under the steep field gradients that develop near breakdown.

#### 3.4.3 Impact Ionization

Avalanche breakdown is modeled using the Selberherr impact ionization model with GaN-specific electron and hole ionization coefficients (`IMPACT selb material=GaN`):

```
an1=an2=2.98e8*3.7,  bn1=bn2=3.44e7
ap1=ap2=2.23e7,      bp1=bp2=2.7e8
```

These coefficients are calibrated to experimental GaN impact ionization data and govern the rate at which high-field carriers generate additional electron-hole pairs, ultimately triggering the current runaway that defines the breakdown voltage [33]. This model is activated exclusively in the breakdown simulation, where lattice temperature coupling (`lat.temp`) and the intrinsic Fermi statistics (`ni.fermi`) are also enabled to capture self-heating effects at high drain voltages.

#### 3.4.4 Ohmic Contacts and Thermal Boundary

Source and drain contacts are defined with a work function of 4.04 eV (`workfunc=4.04`), which matches the electron affinity of GaN and ensures ohmic behavior. The substrate thermal contact (`THERMCONTACT alpha=800`) defines the heat transfer coefficient at the bottom of the substrate, modeling the thermal path to the package heat spreader and providing the boundary condition for self-heating calculations during breakdown simulation.

#### 3.4.5 Numerical Solvers

Two solver configurations are used. The initial direct current (DC) and capacitance-voltage (C-V) simulations use the Newton solver (`METHOD newton climit=1e-4`), which is sufficient for well-conditioned bias sweeps. The breakdown simulation switches to the block iterative solver (`METHOD block`) with significantly tightened convergence tolerances (`ir.tol=1e-35`, `ix.tol=1e-35`, `cr.tol=1e-20`, `cx.tol=1e-20`) to maintain numerical stability as the device approaches the singular point of avalanche breakdown.

### 3.5 Electrical Characterization Methodology

Every device variant in the DoE sweep is subjected to an identical, standardized simulation sequence covering four categories of electrical characteristics. This automated pipeline enables a direct, rigorous, like-for-like comparison across the full parametric sweep. The specific extraction criteria for these key figures of merit are defined below.

#### 3.5.1 Transfer Characteristics and Threshold Voltage ( $V_{th}$ )

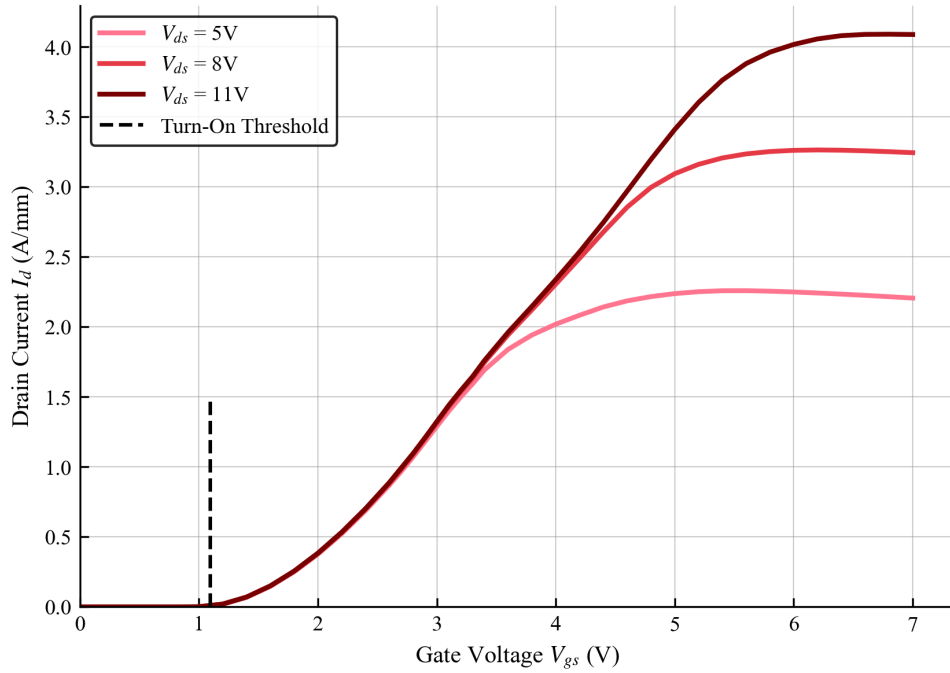


Figure 3.8: Transfer characteristics and threshold voltage extraction using the constant-current criterion.

The transfer characteristics are obtained by sweeping the gate-to-source voltage ( $V_{gs}$ ) from 0 V to 7 V at fixed drain biases of  $V_{ds} = 5$  V, 8 V, and 11 V. To ensure consistency across all device geometries, the threshold voltage ( $V_{th}$ ) is defined strictly using a constant-current turn-on criterion. As illustrated in Figure 3.8,  $V_{th}$  is extracted at the exact gate voltage where the drain current surpasses a predefined normalized threshold of 10 mA/mm, marked by the vertical dashed line indicating the turn-on threshold. Accurate  $V_{th}$  extraction is critical, as it confirms that the recessed p-gate successfully depletes the channel to maintain normally-off operation at zero bias.

### 3.5.2 Output Characteristics and Specific On-State Resistance ( $R_{ds,on}$ )

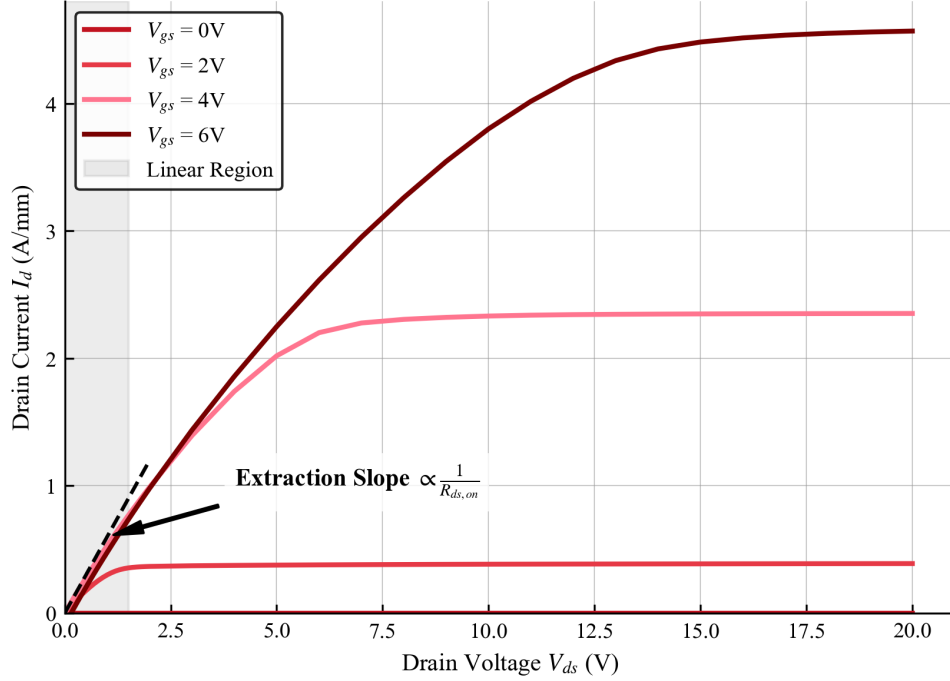


Figure 3.9: Output characteristics highlighting the linear ohmic region and  $R_{ds,on}$  extraction slope.

Output curves ( $I_d$ – $V_{ds}$ ) are generated by sweeping the drain voltage up to 20 V across multiple fixed gate voltage steps ( $V_{gs} = 0$  V to 6 V). The specific on-state resistance ( $R_{ds,on}$ ) is evaluated in the low-voltage linear regime, shaded in gray in Figure 3.9, where the device operates in the ohmic region. The resistance is extracted from the inverse slope of the  $I_d$ – $V_{ds}$  curve at the maximum applied gate drive of  $V_{gs} = 6$  V, ensuring the channel is fully enhanced. Because  $R_{ds,on}$  scales directly with the drift region length, this extraction serves as the primary on-state counterpart to the breakdown voltage measurements, defining the fundamental conduction limits of the  $L_{gd}$  sweep.

### 3.5.3 Capacitance-Voltage (C-V), Charge, and Energy Characteristics

Off-state terminal capacitances are reported by superimposing a small-signal AC perturbation on a high-voltage DC bias sweep at  $V_{gs} = 0$  V. The simulator directly extracts the input ( $C_{iss}$ ), output ( $C_{oss}$ ), and reverse transfer ( $C_{rss}$ ) capacitances. Rather than relying

on a single extraction point, the full highly non-linear capacitance curves are utilized to evaluate the switching performance.

Because the physical structure of the parasitic capacitors changes dynamically as the depletion region expands into the buffer and substrate, the raw capacitance data is mathematically integrated over the applied voltage. These integrations provide standardized, single-value metrics that accurately reflect the large-scale dynamics of the device architecture:

- **Output Energy ( $E_{oss}$ ) and Energy-Related Capacitance ( $C_{o(er)}$ ):** To evaluate large-signal switching losses,  $C_{oss}$  is integrated to compute the total stored output energy:

$$E_{oss} = \int_0^{V_{ds}} V \cdot C_{oss}(V) dV \quad (3.2)$$

From this integral, the energy-related equivalent output capacitance ( $C_{o(er)}$ ) is derived. Defined as the theoretical linear capacitor that stores the identical energy at a given voltage, it is calculated as:

$$C_{o(er)} = \frac{2E_{oss}}{V_{ds}^2} \quad (3.3)$$

- **Output Charge ( $Q_{oss}$ ) and Time-Related Capacitance ( $C_{o(tr)}$ ):** To evaluate switching transition times and dead-time limits,  $C_{oss}$  is integrated to compute the total output charge:

$$Q_{oss} = \int_0^{V_{ds}} C_{oss}(V) dV \quad (3.4)$$

The time-related equivalent output capacitance ( $C_{o(tr)}$ ) is defined as a linear capacitor that requires the exact same charging time:

$$C_{o(tr)} = \frac{Q_{oss}}{V_{ds}} \quad (3.5)$$

- **Gate-to-Drain Charge ( $Q_{gd}$ ):** Finally, the reverse transfer capacitance ( $C_{rss}$ ), which heavily dictates the Miller plateau duration and total gate-drive requirements, is integrated to quantify the total gate-to-drain charge:

$$Q_{gd} = \int_0^{V_{ds}} C_{rss}(V) dV \quad (3.6)$$

### 3.5.4 Breakdown Characteristics and Failure Mechanisms

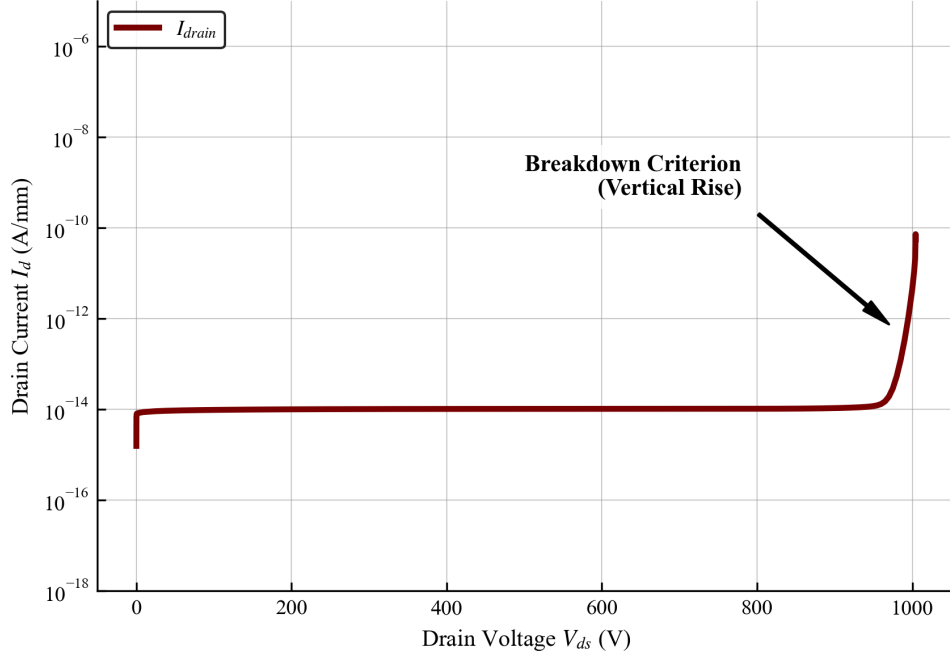


Figure 3.10: Breakdown voltage extraction criterion defined by the sharp vertical rise in off-state drain current.

The breakdown voltage ( $V_{br}$ ) is evaluated by biasing the device deep into the off-state ( $V_{gs} = 0$  V) and ramping the drain voltage until catastrophic leakage occurs. As depicted in Figure 3.10, the drain current remains exceptionally low (on the order of  $10^{-14}$  A/mm) across the majority of the voltage blocking range. The device failure criterion is defined by a sharp, vertical rise in the drain current, visually indicated in the semilogarithmic plot. This vertical wall indicates that the internal electric field has either triggered avalanche multiplication via impact ionization or severely breached the dielectric isolation of the buffer and substrate stack. The voltage precisely at the onset of this vertical rise is recorded as  $V_{br}$  for the specific simulated variant.

## 4 Results

This chapter presents the comprehensive simulation results and physical analysis of the lateral GaN HEMT DoE sweep. The chapter is structured into three main investigative sections: the evaluation of the incumbent silicon substrate baseline, the performance impact of substituting an electrically insulating sapphire substrate, and the electrostatic analysis of field-plate architectures.

### 4.1 Silicon Baseline Performance

This section presents the electrical performance of the lateral GaN HEMT on the silicon substrate platform across all parametric combinations defined in the DoE sweep. Silicon represents the incumbent substrate technology for lateral GaN power devices, offering cost advantages and broad compatibility with established semiconductor fabrication infrastructure. The results establish the performance characteristics of the silicon baseline and identify the fundamental limitations that motivate the substrate optimizations.

### 4.1.1 On-State Resistance

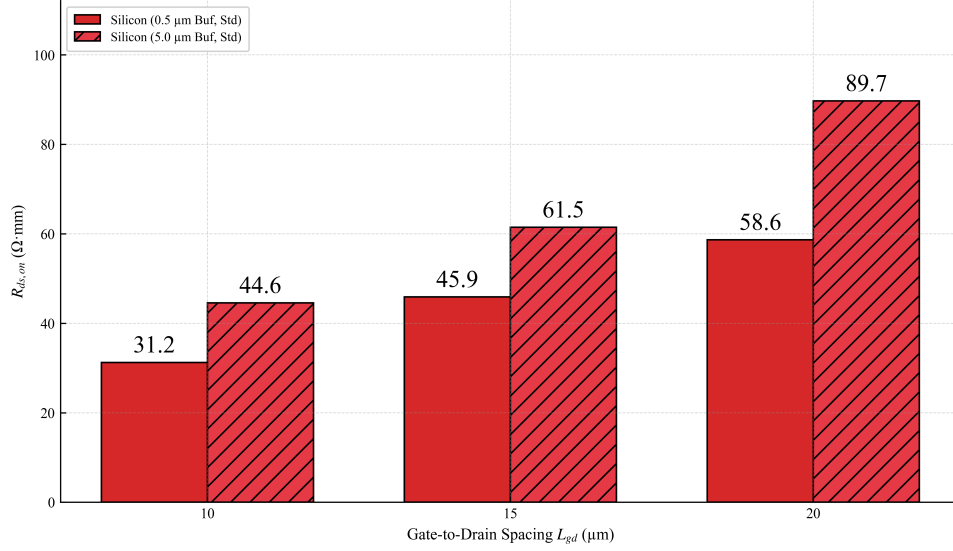


Figure 4.1: Extracted on-state resistance ( $R_{ds,on}$ ) for the silicon baseline, grouped by buffer thickness and gate-to-drain spacing ( $L_{gd}$ ). Thicker buffer layers consistently inflate resistance due to trap-mediated depletion in the access region.

As illustrated in Figure 4.1, on-state resistance ( $R_{ds,on}$ ) on the silicon platform exhibits a dependence on both gate-to-drain spacing ( $L_{gd}$ ) and buffer thickness ( $t_{buf}$ ). It converges to approximately  $31.2 \Omega\cdot\text{mm}$  at  $L_{gd} = 10 \mu\text{m}$  with  $t_{buf} = 0.5 \mu\text{m}$ ,  $44.6 \Omega\cdot\text{mm}$  at  $L_{gd} = 10 \mu\text{m}$  with  $t_{buf} = 5.0 \mu\text{m}$ ,  $45.9 \Omega\cdot\text{mm}$  at  $L_{gd} = 15 \mu\text{m}$  with  $t_{buf} = 0.5 \mu\text{m}$ ,  $61.5 \Omega\cdot\text{mm}$  at  $L_{gd} = 15 \mu\text{m}$  with  $t_{buf} = 5.0 \mu\text{m}$ ,  $58.6 \Omega\cdot\text{mm}$  at  $L_{gd} = 20 \mu\text{m}$  with  $t_{buf} = 0.5 \mu\text{m}$ , and  $89.7 \Omega\cdot\text{mm}$  at  $L_{gd} = 20 \mu\text{m}$  with  $t_{buf} = 5.0 \mu\text{m}$ . Within each  $L_{gd}$  and buffer thickness group, the variation in  $R_{ds,on}$  across substrate thickness combinations remains below 0.3%, confirming that substrate thickness does not contribute measurably to the in-plane conduction path.

The dependence of  $R_{ds,on}$  on buffer thickness is physically attributed to the back-gating effect. In a GaN HEMT grown on a conductive silicon substrate, the substrate potential modulates the electrostatic environment of the GaN buffer layer, which in turn influences the 2DEG charge density in the access region between the gate and drain contacts. A thicker buffer layer increases the vertical separation between the channel and the conductive substrate, partially screening this back-gating modulation and altering the effective sheet resistance of the channel. Furthermore, the acceptor trap distribution within the GaN buffer, positioned 1.0 eV below the conduction band as defined in the methodology, partially compensates the channel charge density in the access region. The combined effect of back-gating modulation and trap-mediated access region depletion [15], [16] produces the observed buffer thickness dependence of  $R_{ds,on}$ , which is therefore a physical consequence of the device architecture rather than a simulation artifact.

## 4.1.2 Breakdown Voltage

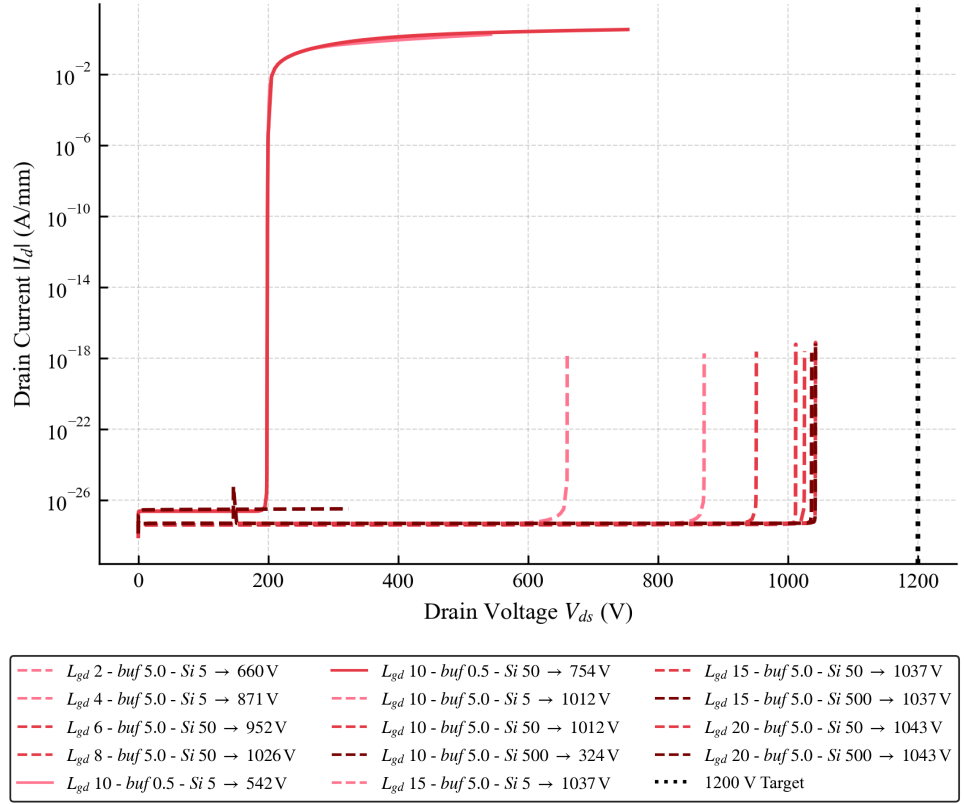


Figure 4.2: Off-state leakage trajectories and breakdown limits for the silicon baseline compared to the 1200 V target. All configurations experience premature vertical punch-through and completely fail to reach the 1200 V qualification threshold.

The breakdown voltage ( $V_{br}$ ) of the silicon baseline configurations is a central finding of this chapter. Figure 4.2 plots the off-state leakage trajectories, demonstrating the abrupt onset of avalanche current that defines device failure. Table 4.1 presents the full  $V_{br}$  results alongside the qualification status of each configuration relative to the 1200 V target class.

Table 4.1: Extracted parameters and qualification status for the silicon baseline.

| $L_{gd}$ ( $\mu\text{m}$ ) | $t_{buf}$ ( $\mu\text{m}$ ) | $t_{sub}$ ( $\mu\text{m}$ ) | $R_{ds,on}$ ( $\Omega\cdot\text{mm}$ ) | $V_{br}$ (V) | % of 1200 V Target | Status        |
|----------------------------|-----------------------------|-----------------------------|----------------------------------------|--------------|--------------------|---------------|
| 10                         | 0.5                         | 5                           | 31.2                                   | 542          | 45.2%              | Non-compliant |
| 10                         | 0.5                         | 50                          | 31.2                                   | 754          | 62.8%              | Non-compliant |
| 10                         | 5.0                         | 5                           | 44.6                                   | 1012         | 84.3%              | Non-compliant |
| 10                         | 5.0                         | 50                          | 44.6                                   | 1012         | 84.3%              | Non-compliant |
| 10                         | 5.0                         | 500                         | 44.6                                   | 324          | 27.0%              | Non-compliant |
| 15                         | 5.0                         | 5                           | 61.5                                   | 1037         | 86.4%              | Non-compliant |
| 15                         | 5.0                         | 50                          | 61.5                                   | 1037         | 86.4%              | Non-compliant |
| 15                         | 5.0                         | 500                         | 61.5                                   | 1037         | 86.4%              | Non-compliant |
| 20                         | 5.0                         | 50                          | 89.7                                   | 1043         | 86.9%              | Non-compliant |
| 20                         | 5.0                         | 500                         | 89.7                                   | 1043         | 86.9%              | Non-compliant |

No silicon configuration satisfies the 1200 V qualification threshold. The maximum  $V_{br}$  recorded across the entire silicon sweep is 1043 V, achieved at  $L_{gd} = 20\ \mu\text{m}$  with a  $5.0\ \mu\text{m}$  buffer, corresponding to 86.9% of the 1200 V target. The majority of configurations fall considerably further short, with several achieving less than 65% of the required blocking voltage.

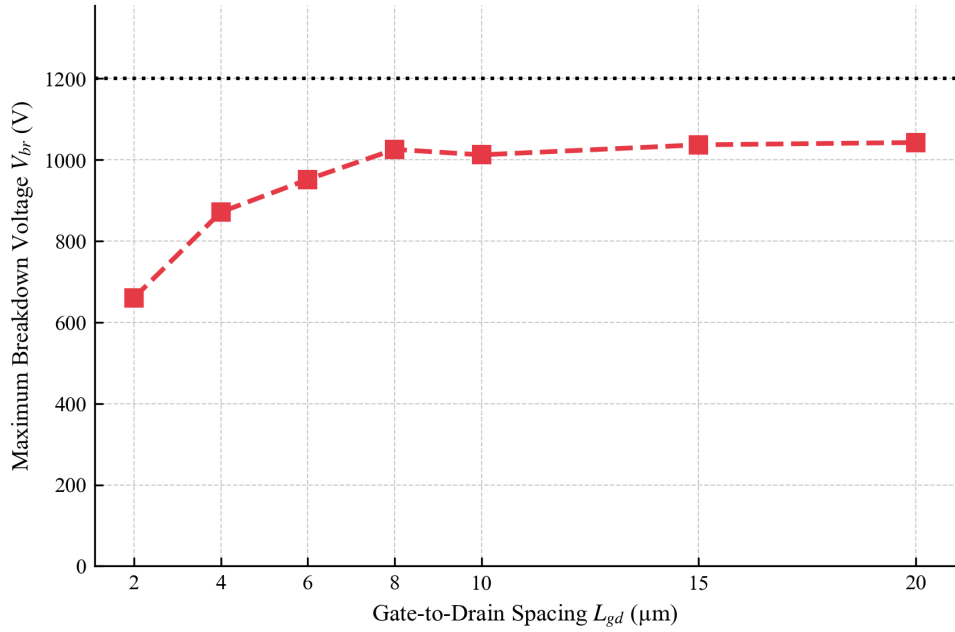


Figure 4.3: Breakdown voltage ( $V_{br}$ ) saturation as a function of gate-to-drain spacing ( $L_{gd}$ ) on the silicon baseline. Voltage scaling abruptly halts beyond an  $8\ \mu\text{m}$  spacing as vertical substrate leakage becomes the dominant failure mechanism.

The visual evidence presented in Figure 4.3 reveals two distinct regimes in the scaling of breakdown voltage ( $V_{br}$ ) with gate-to-drain spacing ( $L_{gd}$ ) on the silicon platform. For shorter drift regions between  $L_{gd} = 2\ \mu\text{m}$  and  $L_{gd} = 8\ \mu\text{m}$ ,  $V_{br}$  exhibits a strong, nearly linear increase, scaling proportionally from 660 V up to a peak of 1025 V. In this initial regime, the breakdown mechanism is laterally limited by the physical distance separating the gate and drain contacts. However, as  $L_{gd}$  is extended beyond  $8\ \mu\text{m}$ , this linear scaling abruptly ceases. At a fixed buffer thickness of  $5.0\ \mu\text{m}$ ,  $V_{br}$  sharply saturates, dropping slightly to 1012 V at  $L_{gd} = 10\ \mu\text{m}$  before plateauing at 1036 V at  $L_{gd} = 15\ \mu\text{m}$  and 1042 V at  $L_{gd} = 20\ \mu\text{m}$ . This represents a negligible total gain of only 30 V across a massive  $10\ \mu\text{m}$  extension of the drift region. This near-complete saturation of  $V_{br}$  with increasing  $L_{gd}$  indicates that the lateral field distribution in the drift region is no longer the governing factor in breakdown onset. Rather, the transition to saturation occurs because vertical current flow through the GaN buffer toward the conductive silicon substrate intercepts the device, establishing a dominant failure mechanism that is entirely insensitive to lateral device geometry. As a consequence, extending  $L_{gd}$  cannot meaningfully raise  $V_{br}$  beyond this substrate-induced ceiling on the silicon platform, rendering the conventional voltage-scaling strategy of increasing drift region length ineffective for reaching the 1200 V threshold.

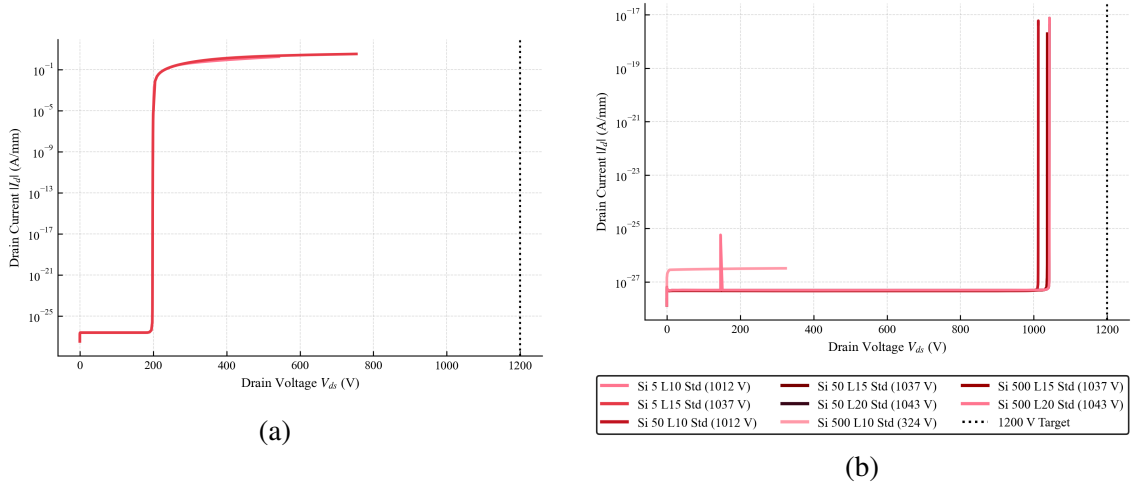


Figure 4.4: Silicon baseline breakdown characteristics evaluated at buffer thicknesses of (a)  $0.5\ \mu\text{m}$  and (b)  $5.0\ \mu\text{m}$ . Increasing the buffer layer depth improves blocking capability but remains insufficient to overcome the conductive substrate limitation.

Buffer thickness is the most influential design variable for  $V_{br}$  on the silicon platform. As shown in Figure 4.4, at  $L_{gd} = 10\ \mu\text{m}$ , reducing the buffer from  $5.0\ \mu\text{m}$  to  $0.5\ \mu\text{m}$  causes  $V_{br}$  to collapse from 1012 V to 542 V at  $5\ \mu\text{m}$  substrate thickness, and from 1012 V to 754 V at  $50\ \mu\text{m}$  substrate thickness. These results confirm that on the silicon platform, the GaN buffer layer is the primary dielectric barrier in the vertical conduction path, and any reduction in buffer thickness directly erodes the blocking capability of the device. Even at

the maximum buffer thickness evaluated in this study, 5.0  $\mu\text{m}$ , the silicon platform fails to reach the 1200 V target, indicating that buffer thickness optimization alone is insufficient to resolve the fundamental voltage limitation of this substrate platform.

On the silicon platform, substrate thickness exerts no meaningful influence on  $V_{br}$  when buffer thickness is held constant. At  $L_{gd} = 15\ \mu\text{m}$  and  $L_{gd} = 20\ \mu\text{m}$  with a 5.0  $\mu\text{m}$  buffer,  $V_{br}$  is identical across the different substrate thickness variants within each configuration. This result is physically consistent with the vertical breakdown mechanism. The conductive silicon body acts as an equipotential surface [13] rather than a dielectric layer, so increasing its depth provides no additional vertical isolation. The blocking voltage is determined by the GaN buffer alone, irrespective of substrate thickness, because the silicon substrate does not contribute to the vertical electric field distribution in any beneficial manner.

### 4.1.3 Output Capacitance and Silicon Limitations

The output capacitance of the silicon baseline configurations reflects the structural constraint identified in the breakdown analysis. Since the conductive silicon body acts as an equipotential surface, it effectively reduces the capacitor plate separation to the GaN buffer thickness alone, producing output capacitance values that are substantially higher than those achievable on an insulating substrate at equivalent geometries. Substrate thickness on silicon exerts only a marginal effect on capacitance, which is attributable to the finite resistivity of the silicon substrate under high reverse bias conditions rather than to any genuine dielectric isolation mechanism.

The silicon baseline sweep establishes two compounding performance limitations that prevent the device from reaching the 1200 V target class. First, the conductive silicon substrate imposes a hard ceiling on  $V_{br}$  by providing a direct vertical current path that bypasses the lateral drift region. This ceiling cannot be raised by extending  $L_{gd}$ , by increasing substrate thickness, or by any other purely lateral or vertical geometric adjustment within the silicon platform. Second, the same conductive substrate eliminates the substrate contribution as a dielectric layer, confining the effective plate separation of the output capacitance to the GaN buffer thickness alone. Addressing either limitation requires a substrate material change rather than incremental adjustments to the GaN epitaxial stack or device lateral geometry.

## 4.2 Sapphire Substrate Evaluation

This section presents the electrical performance of the lateral GaN HEMT with sapphire substituted as the substrate material, evaluated across the same parametric sweep. Sapphire is an electrically insulating, wide-bandgap oxide with high dielectric strength, properties that directly address the two fundamental limitations identified in the silicon baseline.

### 4.2.1 On-State Resistance

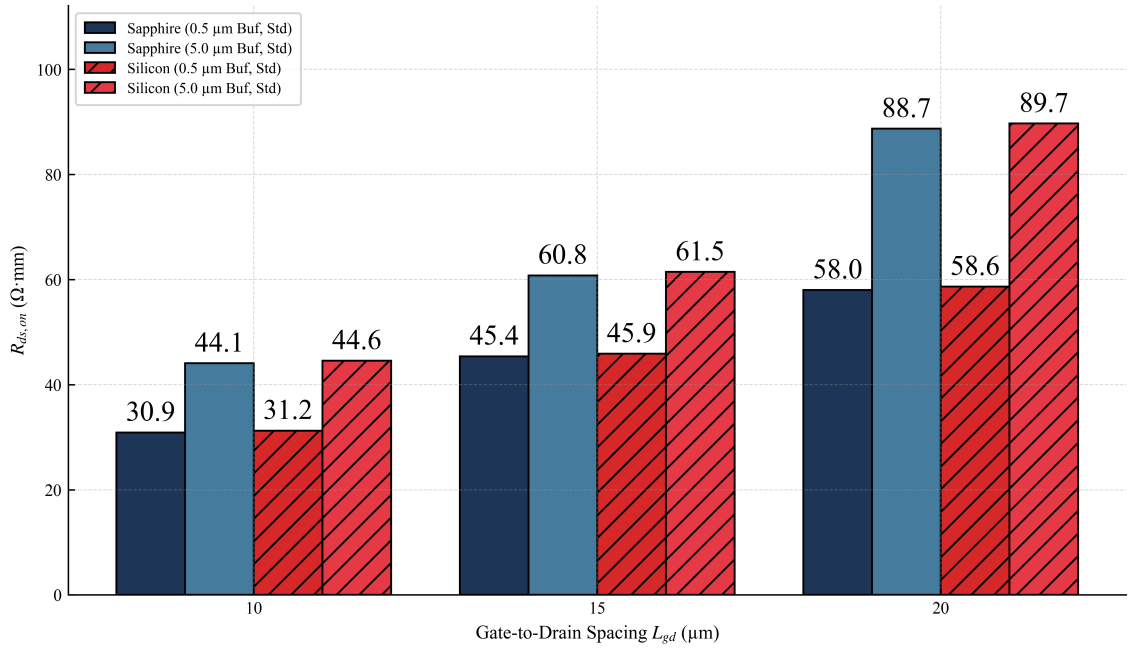


Figure 4.5: Bar Chart of on-state resistance ( $R_{ds,on}$ ) scaling with gate-to-drain spacing ( $L_{gd}$ ) across the silicon and sapphire platforms. The insulating substrate preserves identical in-plane conduction performance while maintaining a strictly linear geometric scaling.

On-state resistance ( $R_{ds,on}$ ) on the sapphire platform exhibits a direct linear scaling relationship with gate-to-drain spacing ( $L_{gd}$ ), as illustrated in Figure 4.5. As the lateral drift region length increases from 10  $\mu\text{m}$  to 15  $\mu\text{m}$  and 20  $\mu\text{m}$ , the conduction path lengthens proportionally, driving a steady and predictable linear increase in total channel resistance.

Specifically, for the 0.5  $\mu\text{m}$  buffer sapphire variant,  $R_{ds,on}$  starts at approximately 30.9  $\Omega\cdot\text{mm}$  at  $L_{gd} = 10 \mu\text{m}$ , rises to 45.4  $\Omega\cdot\text{mm}$  at  $L_{gd} = 15 \mu\text{m}$ , and reaches 58.0  $\Omega\cdot\text{mm}$

at  $L_{gd} = 20\ \mu\text{m}$ . For the thicker  $5.0\ \mu\text{m}$  buffer variant, resistance scales from  $44.1\ \Omega\cdot\text{mm}$  to  $60.8\ \Omega\cdot\text{mm}$  and  $88.7\ \Omega\cdot\text{mm}$  across the same respective spacings.

Furthermore, comparing the sapphire data directly against the silicon baseline reveals nearly identical resistance values across all equivalent configurations (for instance,  $30.9\ \Omega\cdot\text{mm}$  on sapphire versus  $31.2\ \Omega\cdot\text{mm}$  on silicon at  $L_{gd} = 10\ \mu\text{m}$  with a  $0.5\ \mu\text{m}$  buffer). This close alignment confirms that substituting the substrate material does not alter the in-plane conduction path or the 2DEG sheet density.

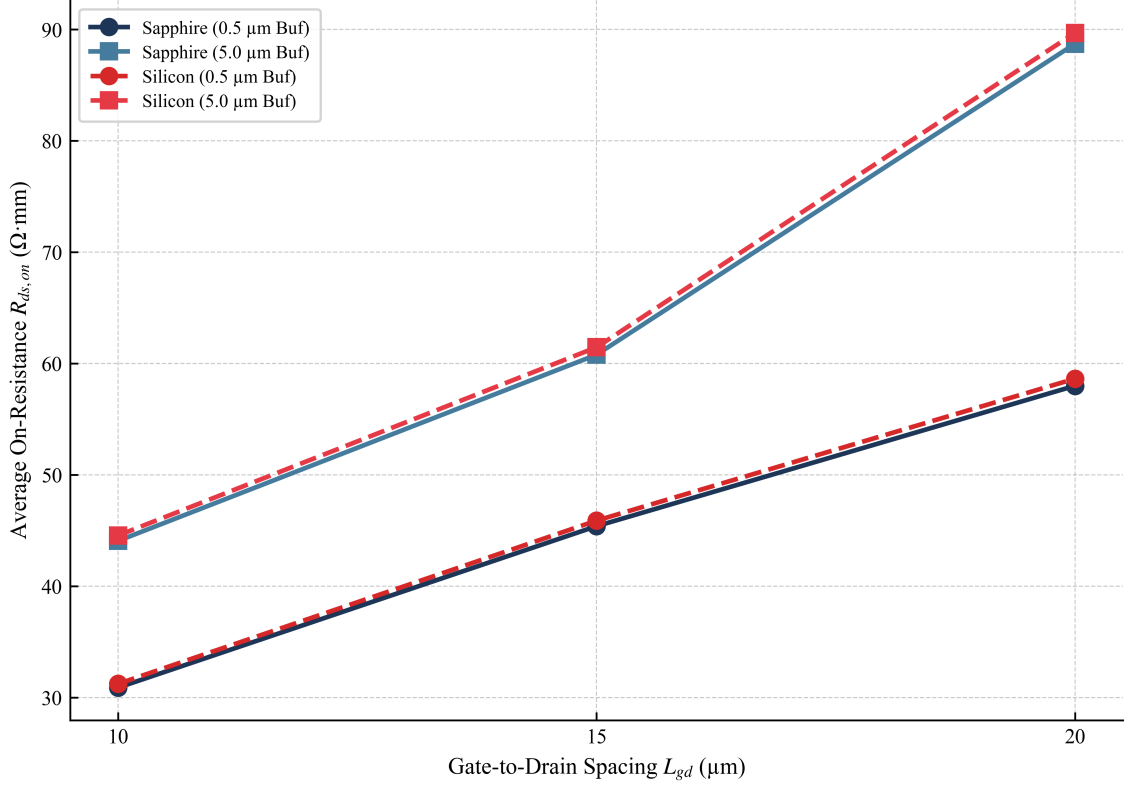


Figure 4.6: Extracted on-state resistance ( $R_{ds,on}$ ) as a function of lateral gate-to-drain spacing ( $L_{gd}$ ) across the silicon and sapphire platforms.

The primary divergence in on-state resistance is instead governed by buffer thickness. Across both substrate platforms, devices utilizing the  $5.0\ \mu\text{m}$  buffer consistently exhibit higher resistance than those with the  $0.5\ \mu\text{m}$  buffer. This penalty is driven by trap-mediated access region depletion, where a thicker carbon-doped buffer intercepts a greater fraction of vertical field lines [14], [15] and depletes the active channel charge density.

### 4.2.2 Breakdown Voltage and Substrate Scaling

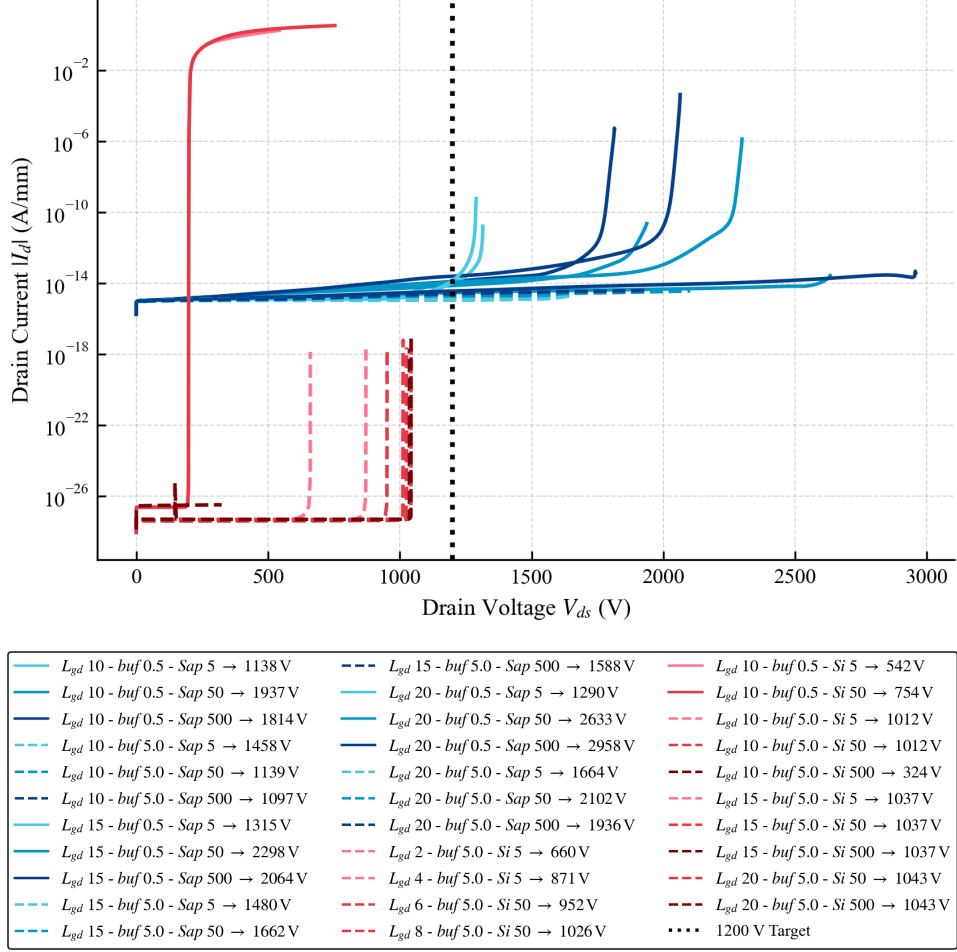


Figure 4.7: Comparison of off-state leakage and breakdown characteristics ( $I_{ds}$  vs.  $V_{ds}$ ) across the silicon and sapphire platforms. Eliminating the conductive back-plane enables the sapphire architectures to comfortably surpass the 1200 V target.

The substitution of the silicon substrate with an electrically insulating sapphire platform results in a massive improvement in breakdown voltage performance. As demonstrated in Figure 4.7, the blocking voltage expands well beyond the 1200 V target line, reaching up to nearly 3000 V depending on the chosen geometry. Because sapphire is an electrical insulator, it completely suppresses the vertical punch-through leakage path that prematurely caps breakdown on conductive silicon substrates. Consequently, the breakdown voltage becomes genuinely sensitive to lateral geometry, allowing the device to fully leverage the drift region.

## 4 Results

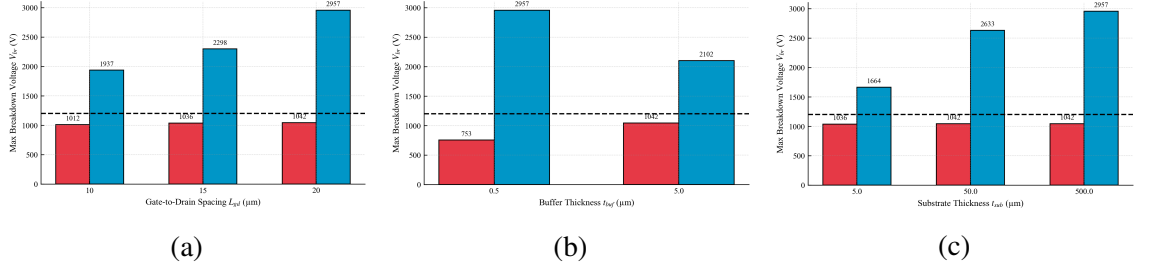


Figure 4.8: Parametric evaluation of breakdown voltage on the sapphire (Blue) platform depicting the impact of (a) gate-to-drain spacing ( $L_{gd}$ ), (b) buffer thickness, and (c) substrate thickness when compared to Silicon (Red).

As highlighted in Figure 4.8a, extending the gate-to-drain spacing ( $L_{gd}$ ) on sapphire produces a strong and consistent increase in breakdown voltage. At a 50  $\mu\text{m}$  substrate thickness and a 0.5  $\mu\text{m}$  buffer,  $V_{br}$  scales from 1937 V at  $L_{gd} = 10 \mu\text{m}$  up to 2633 V at  $L_{gd} = 20 \mu\text{m}$ . This proves that the sapphire substrate successfully restores lateral drift length as an effective voltage-scaling lever.

Beyond electrical behavior, the parametric trends regarding buffer and substrate thicknesses carry critical manufacturing and economic implications. Regarding epitaxial design, a thinner buffer layer (0.5  $\mu\text{m}$ ) combined with sapphire delivers exceptional electrical performance while offering massive manufacturing time and cost savings. Growing thick carbon-doped GaN buffer layers via MOCVD requires extensive reactor time and high thermal budgets. Reducing the buffer thickness from 5.0  $\mu\text{m}$  to 0.5  $\mu\text{m}$  drastically cuts reactor run times, lowers precursor consumption, and minimizes parasitic wall depositions, thereby reducing chamber cleaning downtime and production costs in high-volume manufacturing.

Similarly, as shown in Figure 4.8c, increasing the sapphire substrate thickness yields higher breakdown voltages by deepening the total dielectric isolation stack. From a fabrication standpoint, utilizing a 500  $\mu\text{m}$  substrate offers a major economic advantage because it corresponds to the standard as-grown wafer thickness for commercially available sapphire substrates. This completely eliminates the need for post-growth wafer thinning, grinding, or chemical-mechanical polishing, avoiding additional yield losses, process complexity, and manufacturing expense. The 500  $\mu\text{m}$  substrate therefore provides maximum dielectric isolation depth natively, requiring no extra processing steps beyond a standard sapphire process flow.

### 4.2.3 Electrostatic Potential and Electric Field Distribution

To elucidate the physical mechanisms driving the superior breakdown performance of the sapphire platform, two-dimensional electrostatic simulations of potential and electric field distributions were evaluated under high reverse bias conditions.

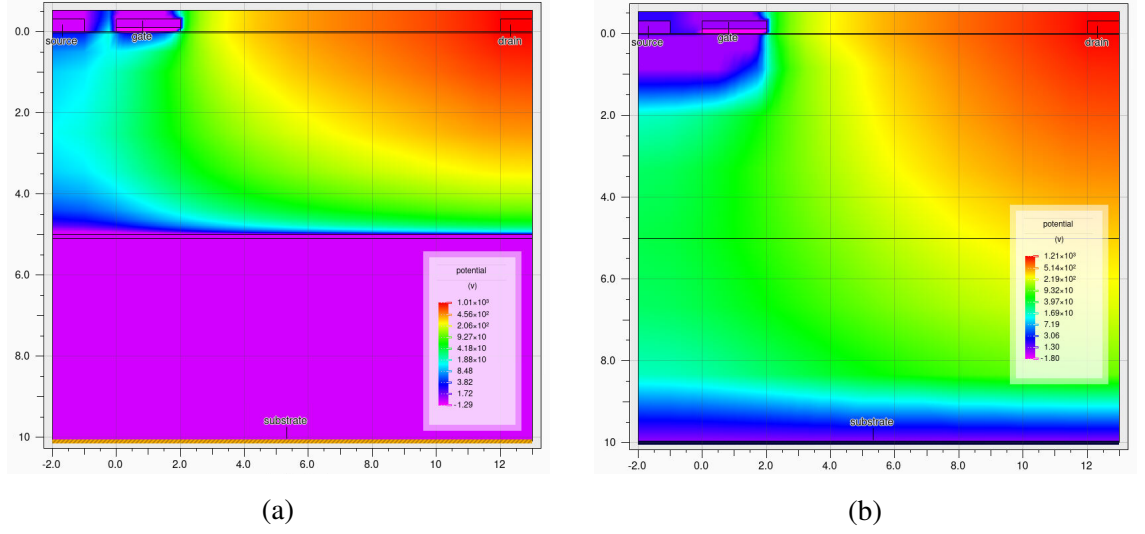


Figure 4.9: Steady-state electrostatic potential distribution comparing (a) the silicon baseline and (b) the sapphire substrate platform. Equipotential lines severely compress against the silicon backplane but distribute smoothly through the full depth of the sapphire dielectric.

As illustrated in the steady-state potential contours of Figure 4.9, the choice of substrate fundamentally alters the internal electrostatic landscape of the transistor. In the silicon baseline configuration, the conductive silicon substrate acts as a fixed equipotential plane [13], forcing the potential lines to compress and crowd tightly near the bottom of the GaN buffer layer. This severe potential crowding results in an abnormally thin effective voltage-blocking region, concentrating almost the entire drain voltage drop across the vertical hetero-epitaxial stack. Conversely, the electrically insulating sapphire substrate eliminates this fixed ground plane limitation. The potential contours in Figure 4.9b demonstrate a smooth, gradual spreading of equipotential lines throughout the entire depth of the sapphire substrate. This behavior confirms that sapphire acts as an active dielectric layer that shares the high-voltage burden rather than forcing a localized electrostatic collapse.

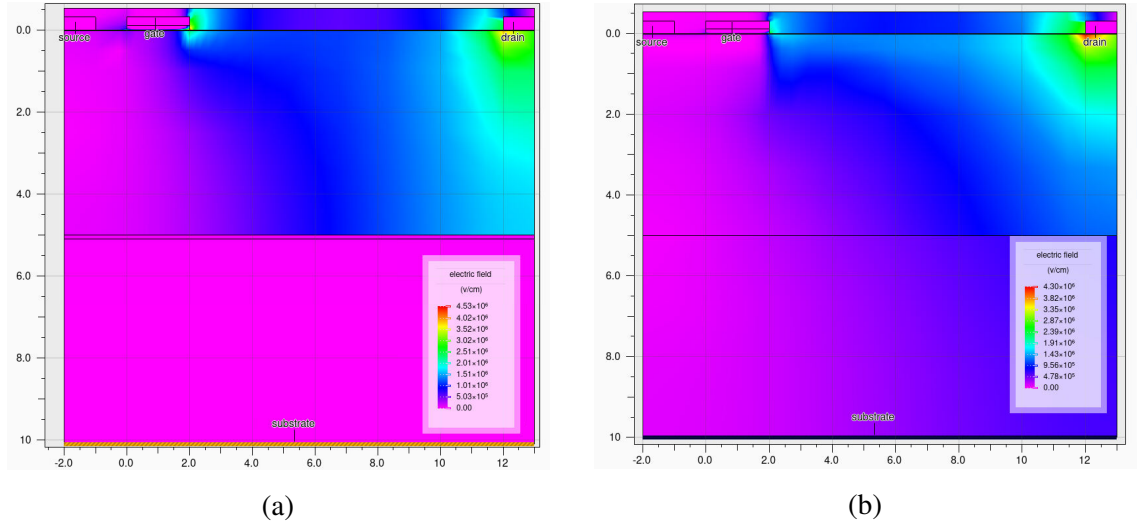


Figure 4.10: Two-dimensional electric field distribution comparing (a) the silicon substrate and (b) the sapphire substrate platforms. The sapphire platform successfully suppresses vertical field crowding and shifts the electrostatic stress back to the surface drift region.

The corresponding two-dimensional electric field distributions shown in Figure 4.10 provide direct visual proof of why vertical punch-through failure occurs on silicon. On the silicon platform, high electric field intensities concentrate sharply at the buffer-substrate interface, creating a vulnerable leakage path that triggers premature avalanche breakdown before the lateral drift region can be fully utilized. The sapphire platform successfully suppresses vertical field crowding, redistributing the electrostatic stress more uniformly and shifting the primary electric field management back to the lateral surface drift region. Consequently, the device avoids destructive vertical leakage, unlocking the high breakdown voltages observed across the sapphire design sweep.

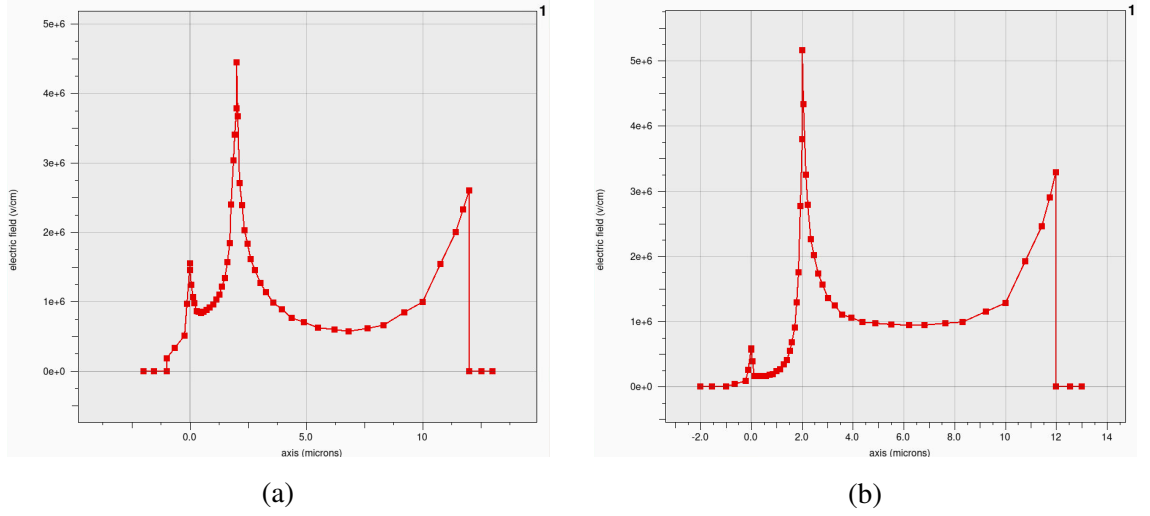
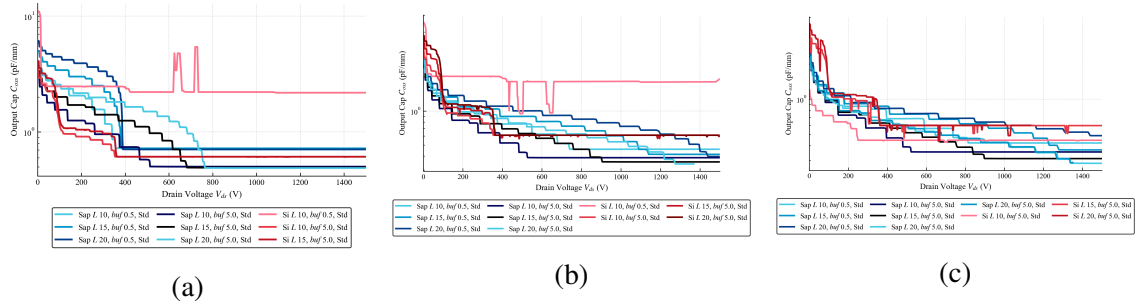


Figure 4.11: One-dimensional electric field distributions along the active channel prior to breakdown for (a) the silicon baseline and (b) the sapphire baseline. Both unplated architectures exhibit a field peak at the drain-side gate edge that drives premature avalanche breakdown.

To physically understand the lateral breakdown limitations once vertical failure is suppressed, it is necessary to examine the one-dimensional internal electrostatic profile along the active channel. In conventional, unplated AlGaIn/GaN HEMTs, the lateral voltage drop is not distributed evenly across the drift region. Instead, it concentrates heavily at the drain-side edge of the gate electrode [8]. The electric field distributions for the unplated silicon and sapphire baselines just prior to breakdown are compared in Figure 4.11. Both platforms exhibit the classic electrostatic signature of a lateral HEMT: a massive, highly localized electric field peak precisely at the gate edge ( $x \approx 2 \mu\text{m}$ ). On the silicon platform, this primary gate-edge peak reaches approximately  $4.5 \times 10^6 \text{ V/cm}$ . On the sapphire platform, the peak field is even more aggressive, exceeding  $5.0 \times 10^6 \text{ V/cm}$ .

The higher peak magnitude on the sapphire platform is a direct consequence of eliminating the conductive silicon backplane. Because the sapphire substrate prevents premature vertical punch-through, the device is capable of sustaining higher applied lateral voltages. However, without any top-side electrostatic shaping structures, this increased lateral voltage forces an even steeper potential gradient at the gate edge. While both profiles display a secondary field peak at the drain interface ( $x = 12 \mu\text{m}$ ) due to the depletion region reaching the highly doped ohmic contact, this drain peak remains strictly subordinate to the primary gate-edge spike.

This extreme gate-edge field crowding acts as the fundamental lateral failure mechanism. When the localized electric field exceeds the critical threshold of the semiconductor, it triggers catastrophic avalanche multiplication via impact ionization. These profiles demonstrate that extending the drift region length ( $L_{gd}$ ) alone is insufficient to scale the



## 4 Results

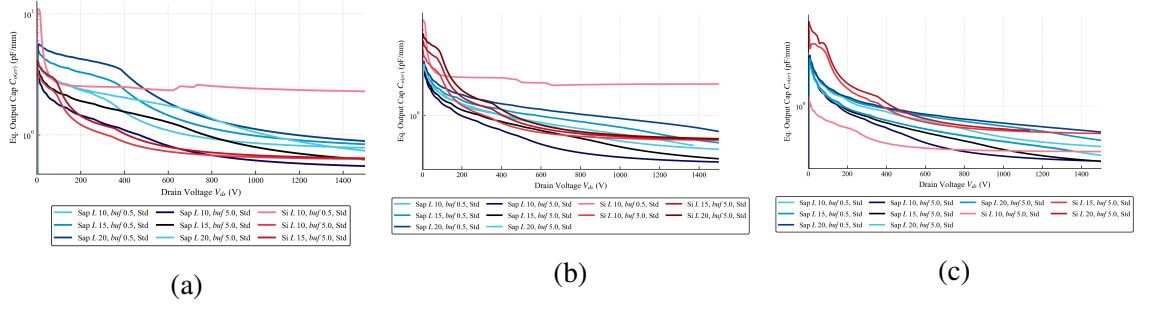


Figure 4.13: Energy-related output capacitance ( $C_{o(er)}$ ) evaluated for sapphire substrate thicknesses of (a) 5.0 μm, (b) 50.0 μm, and (c) 500.0 μm. Thicker sapphire substrates directly minimize the equivalent linear capacitance evaluated at high operating voltages.

This capacitance reduction carries directly into the energy-related output capacitance  $C_{o(er)}$  and stored output energy  $E_{o(er)}$  plotted in Figures 4.13 and 4.14.

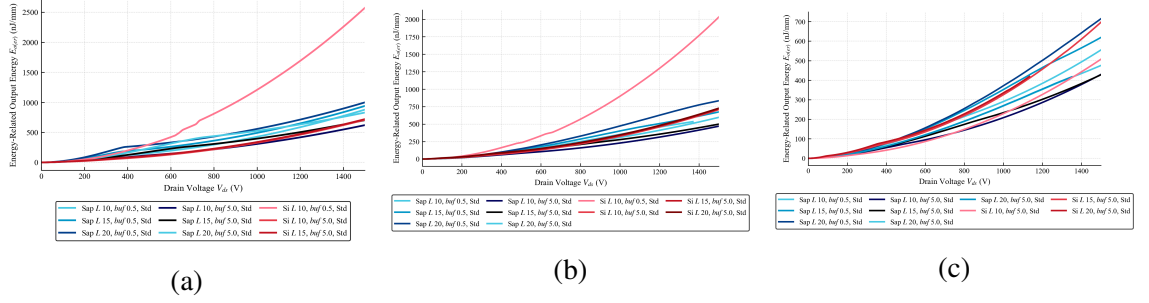


Figure 4.14: Stored output energy ( $E_{o(er)}$ ) evaluated for sapphire substrate thicknesses of (a) 5.0 μm, (b) 50.0 μm, and (c) 500.0 μm.

The thickest 500.0 μm sapphire substrate strictly minimizes stored energy during hard switching events compared to silicon devices, which suffer from elevated energy dissipation due to their conductive substrate grounding.

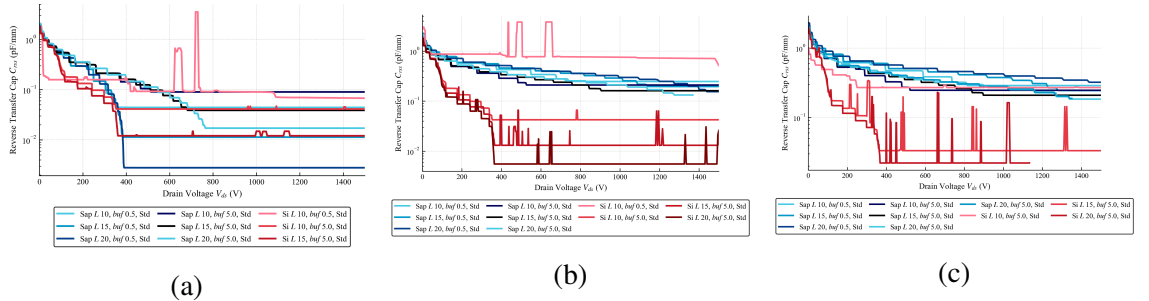


Figure 4.15: Reverse transfer capacitance ( $C_{rss}$ ) evaluated for sapphire substrate thicknesses of (a) 5.0 μm, (b) 50.0 μm, and (c) 500.0 μm. Both Architectures maintain low feedback capacitance across the full applied voltage sweep.

## 4 Results

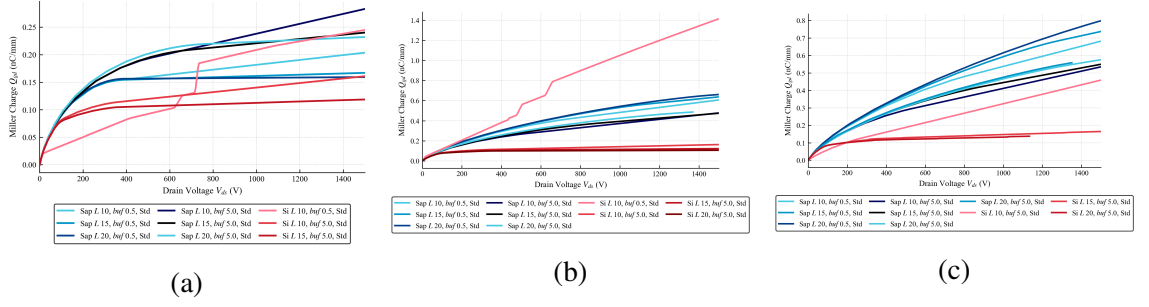


Figure 4.16: Miller charge ( $Q_{gd}$ ) evaluated for sapphire substrate thicknesses of (a) 5.0  $\mu\text{m}$ , (b) 50.0  $\mu\text{m}$ , and (c) 500.0  $\mu\text{m}$ .

Furthermore, as illustrated in Figures 4.15 and 4.16, the reverse transfer capacitance  $C_{rss}$  and Miller charge  $Q_{gd}$  dictate feedback dynamics and  $dv/dt$  immunity. The sapphire platforms exhibit well-controlled feedback characteristics, ensuring superior high-frequency switching stability.

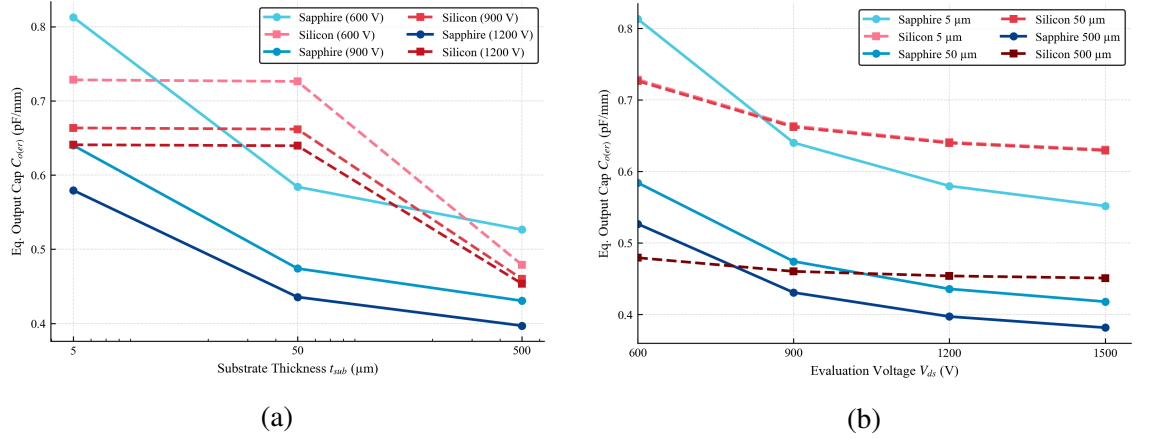


Figure 4.17: Capacitance evaluation comparing the sapphire architecture against the silicon, detailing (a) dielectric scaling trends and (b) dynamic capacitance-voltage profiles. Removing the grounded conductive backplane yields substantial reductions in output capacitance across all bias points.

The capacitance analysis in Figure 4.17 highlights the structural advantage of the insulating platform. By removing the grounded conductive backplane, the sapphire substrate acts as an additional series dielectric layer, fundamentally lowering the output capacitance across the entire voltage sweep.

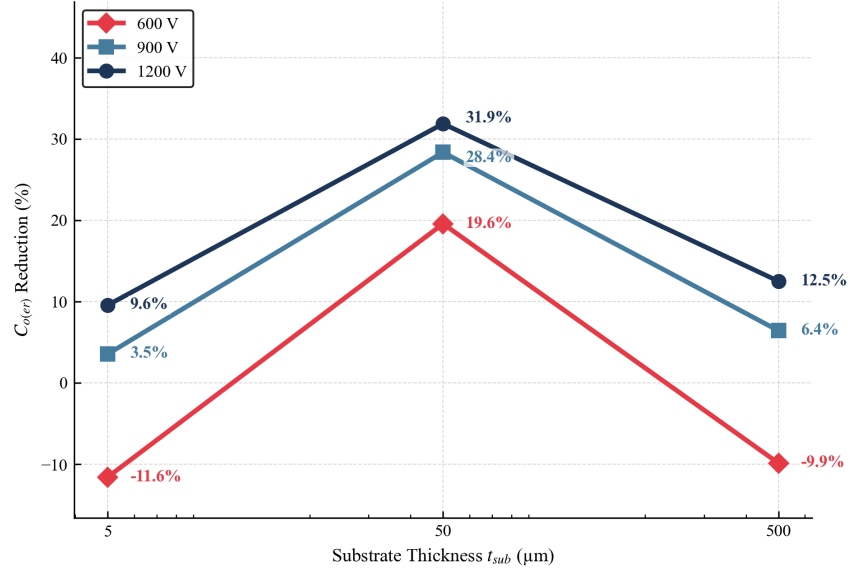


Figure 4.18: Percentage reduction in energy-related output capacitance ( $C_{o(er)}$ ) achieved by the sapphire platform compared to the silicon baseline. The optimized insulating substrate delivers up to a 31.9% improvement in dynamic parasitics at the operating voltage.

As explicitly quantified in Figure 4.18, substituting silicon with sapphire yields a massive reduction in dynamic parasitics. At the 1200 V evaluation bias, the sapphire platform achieves up to a 31.9% reduction in  $C_{o(er)}$  at a 50  $\mu\text{m}$  substrate thickness. This confirms that the insulating substrate provides superior dielectric isolation, which directly translates into lower switching energy dissipation.

#### 4.2.5 Intermediate Voltage Class Benchmarking

To rigorously evaluate the isolated impact of the substrate material, the sapphire and silicon platforms are benchmarked at intermediate 600 V and 900 V evaluation biases using a strictly identical vertical architecture. Table 4.2 compares the performance of both materials across the three lateral drift geometries while holding the epitaxial buffer thickness fixed at 5.0  $\mu\text{m}$  and the substrate thickness fixed at the maximum 500  $\mu\text{m}$  depth.

Table 4.2: Performance comparison of silicon and sapphire platforms using a fixed  $5.0\mu\text{m}$  buffer and  $500\mu\text{m}$  substrate evaluated at 600 V and 900 V operating classes.

| $L_{gd}$<br>( $\mu\text{m}$ ) | Material | $V_{br}$<br>(V) | $R_{ds,on}$<br>( $\Omega\cdot\text{mm}$ ) | 600 V Evaluation    |                                       | 900 V Evaluation    |                                       |
|-------------------------------|----------|-----------------|-------------------------------------------|---------------------|---------------------------------------|---------------------|---------------------------------------|
|                               |          |                 |                                           | $C_{o(er)}$ (pF/mm) | FOM <sup>†</sup> ( $\times 10^{-4}$ ) | $C_{o(er)}$ (pF/mm) | FOM <sup>†</sup> ( $\times 10^{-4}$ ) |
| 10                            | Silicon  | 324             | 44.60                                     | 0.4792              | 356.21                                | 0.4602              | 228.05                                |
|                               | Sapphire | 1097            | 44.08                                     | 0.5265              | 386.80                                | 0.4306              | 210.90                                |
| 15                            | Silicon  | 1037            | 61.50                                     | 0.7542              | 773.06                                | 0.6643              | 453.94                                |
|                               | Sapphire | 1588            | 60.80                                     | 0.6284              | 636.78                                | 0.5037              | 340.28                                |
| 20                            | Silicon  | 1043            | 89.70                                     | 0.8021              | 1199.14                               | 0.6868              | 684.51                                |
|                               | Sapphire | 1936            | 88.75                                     | 0.6673              | 987.05                                | 0.5628              | 554.98                                |

<sup>†</sup> Figure of Merit (FOM) evaluated as  $\frac{R_{ds,on} \cdot C_{o(er)}}{V_{op}}$ , ( $V_{op}$  (Operating Voltage)).

This fixed-geometry comparison exposes the severe electrostatic penalties inherent to the conductive silicon platform. At the shortest  $L_{gd} = 10\mu\text{m}$  spacing, the silicon device catastrophically fails at 324 V due to vertical punch-through, rendering it physically incapable of operating in either the 600 V or 900 V application space. In contrast, the identically dimensioned sapphire architecture provides the necessary dielectric isolation to comfortably sustain 1097 V. This proves that achieving medium-voltage blocking capabilities on a compact lateral footprint is impossible on a grounded conductive backplane.

For the extended  $L_{gd} = 15\mu\text{m}$  and  $20\mu\text{m}$  architectures where the silicon devices successfully survive the intermediate voltage biases, the dynamic switching advantage of sapphire becomes explicit. As the operating voltage ( $V_{op}$ ) increases from 600 V to 900 V, the equivalent energy-related capacitance ( $C_{o(er)}$ ) drops across all configurations due to the physical expansion of the depletion region, which naturally lowers the normalized figure of merit. However, the conductive silicon body acts as a rigid, active parallel capacitor plate located immediately beneath the buffer layer, restricting the effective dielectric thickness and locking  $C_{o(er)}$  at artificially elevated values. The sapphire platform completely circumvents this constraint. By acting as a deep  $500\mu\text{m}$  insulating extension to the buffer, sapphire allows the electric field lines to distribute widely, strictly suppressing the dynamic output capacitance.

This reduction in parasitic capacitive loading directly translates into superior switching efficiency. At  $L_{gd} = 15\mu\text{m}$ , substituting silicon with sapphire lowers the hard-switching figure of merit by 17.6% at 600 V and 25.0% at 900 V, while simultaneously raising the avalanche boundary from 1037 V to 1588 V. At the  $L_{gd} = 20\mu\text{m}$  limit, the sapphire substrate provides an 85.6% gain in blocking voltage (1936 V vs. 1043 V) and reduces the 900 V evaluation figure of merit from 684.51 to 554.98 ( $\times 10^{-4}$ ). Ultimately, the data dictates that to maintain acceptable on-state resistance without sacrificing breakdown voltage or inducing severe high-frequency switching losses, the architectural transition to an insulating platform is required.

### 4.2.6 Consolidated Figure of Merit Ranking and Trade-Off Analysis

Consolidating all eighteen sapphire configurations by their dynamic figure of merit evaluated at the 1200 V target class yields the performance hierarchy presented in Table 4.3. It must be explicitly noted that the silicon baseline is entirely absent from this ranking because every simulated silicon configuration failed to reach the 1200 V threshold due to vertical punch-through, rendering them completely unqualified for high-voltage applications.

Table 4.3: Sapphire configurations ranked by figure of merit at 1200 V (lower values indicate better performance).

| Rank | $L_{gd}$ ( $\mu\text{m}$ ) | $t_{buf}$ ( $\mu\text{m}$ ) | $t_{sub}$ ( $\mu\text{m}$ ) | $R_{ds,on}$ ( $\Omega\cdot\text{mm}$ ) | $V_{br}$ (V) | FOM <sup>†</sup> ( $\times 10^{-4}$ ) |
|------|----------------------------|-----------------------------|-----------------------------|----------------------------------------|--------------|---------------------------------------|
| 1    | 10                         | 0.5                         | 500                         | 30.86                                  | 1814         | 137.00                                |
| 2    | 10                         | 0.5                         | 50                          | 30.87                                  | 1937         | 146.84                                |
| 3    | 15                         | 5.0                         | 500                         | 60.80                                  | 1588         | 212.41                                |
| 4    | 10                         | 5.0                         | 5                           | 44.08                                  | 1458         | 212.88                                |
| 5    | 15                         | 0.5                         | 500                         | 45.43                                  | 2064         | 244.10                                |
| 6    | 15                         | 5.0                         | 50                          | 60.80                                  | 1662         | 250.40                                |
| 7    | 15                         | 0.5                         | 50                          | 45.42                                  | 2298         | 266.45                                |
| 8    | 20                         | 5.0                         | 5                           | 41.95                                  | 1664         | 307.06                                |
| 9    | 20                         | 0.5                         | 500                         | 58.06                                  | 2958         | 336.81                                |
| 10   | 15                         | 0.5                         | 5                           | 45.35                                  | 1315         | 340.76                                |
| 11   | 15                         | 5.0                         | 5                           | 60.79                                  | 1480         | 354.58                                |
| 12   | 20                         | 5.0                         | 500                         | 88.75                                  | 1936         | 364.12                                |
| 13   | 20                         | 0.5                         | 50                          | 58.05                                  | 2633         | 421.34                                |
| 14   | 20                         | 5.0                         | 50                          | 88.73                                  | 2102         | 469.49                                |
| 15   | 20                         | 0.5                         | 5                           | 57.90                                  | 1290         | 477.79                                |

<sup>†</sup> Figure of Merit (FOM) evaluated as  $\frac{R_{ds,on} \cdot C_{o(er)}}{V_{op}}$ .

The tabulated data reveals that the top-tier performance is strictly dominated by configurations utilizing the shortest lateral drift geometry ( $L_{gd} = 10 \mu\text{m}$ ) combined with the thinnest epitaxial buffer ( $t_{buf} = 0.5 \mu\text{m}$ ). The paramount configuration ( $L_{gd} = 10 \mu\text{m}$ ,  $0.5 \mu\text{m}$  buffer,  $500 \mu\text{m}$  substrate) achieves a figure of merit of  $137.00 \times 10^{-4}$  with a breakdown voltage of 1814 V, providing a comfortable electrostatic margin of 51.2% above the 1200 V qualification threshold. Its superior ranking is driven by combining the lowest on-state resistance in the sweep ( $30.86 \Omega\cdot\text{mm}$ , achieved via the thinner buffer which minimizes access region trap depletion) with a minimized output capacitance ( $0.5327 \text{ pF/mm}$ )

enabled by the deep 500  $\mu\text{m}$  sapphire dielectric.

Conversely, configurations employing extended gate-to-drain spacings ( $L_{gd} = 20\mu\text{m}$ ) populate the lowest ranks of the hierarchy, yielding figures of merit exceeding  $300 \times 10^{-4}$ . While these extended geometries successfully raise the breakdown voltage to nearly 3000 V, the linear inflation of in-plane conduction resistance overwhelms any marginal reductions in high-voltage output capacitance, fundamentally degrading the overall switching efficiency. Furthermore, architectures utilizing the thicker 5.0  $\mu\text{m}$  buffer consistently exhibit degraded figures of merit relative to their 0.5  $\mu\text{m}$  counterparts due to the aforementioned trap-mediated channel depletion penalty.

To visually synthesize these broader performance trade-offs across the entire design space, the configurations are mapped onto a high-frequency performance power plot in Figure 4.19.

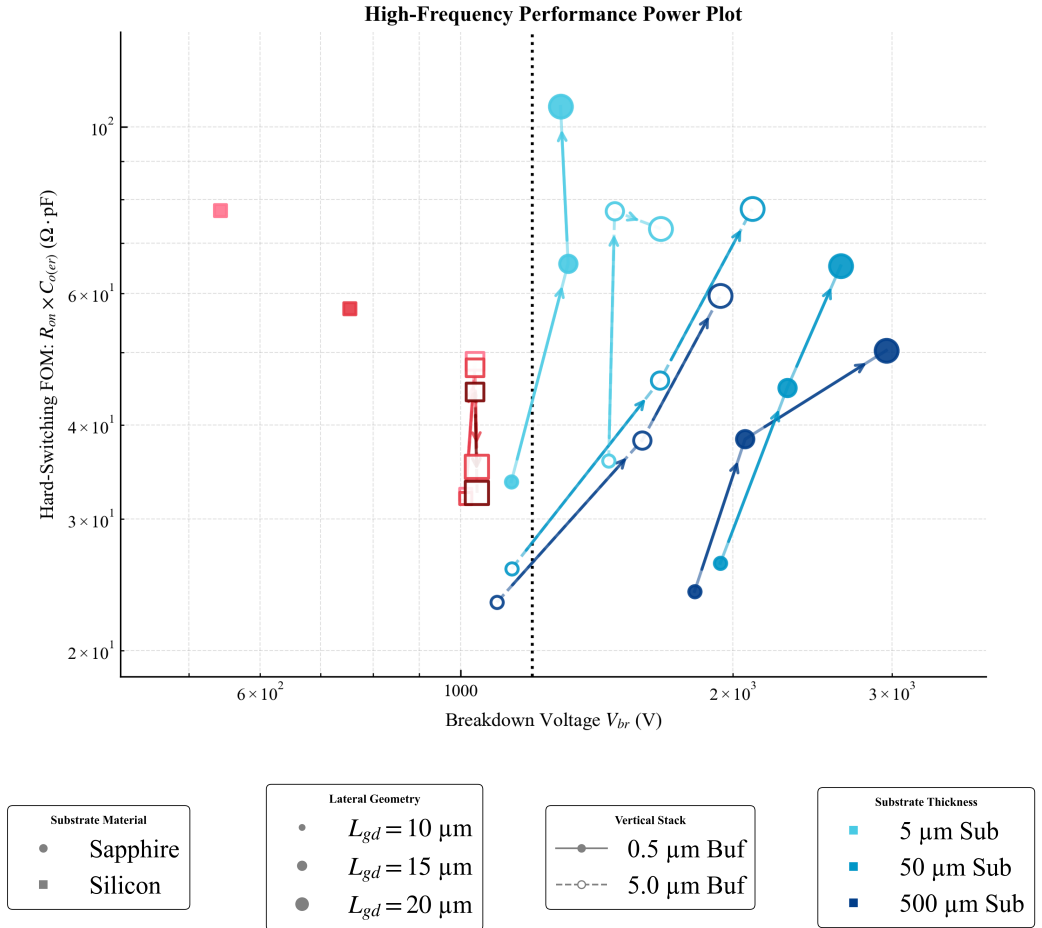


Figure 4.19: High-frequency performance power plot mapping the hard-switching figure of merit ( $R_{ds,on} \times C_{oss}$ ) against breakdown voltage ( $V_{br}$ ). The standalone legend panel details the architectural variables mapped to marker shape, size, color, and fill.

The axes map the hard-switching figure of merit product ( $R_{ds,on} \times C_{o(er)}$ ) on a logarithmic scale against breakdown voltage ( $V_{br}$ ). In this representation, lower values on the vertical axis denote superior dynamic performance and lower energy losses. To evaluate the intersecting physical parameters, the visual markers systematically encode four distinct architectural variables into the two-dimensional space. The shape of the marker identifies the substrate material, with circles representing sapphire and squares representing silicon. The base color of the trajectory corresponds to the substrate thickness, ranging from light colors for 5.0  $\mu\text{m}$  variants to dark colors for 500.0  $\mu\text{m}$  variants. The vertical epitaxial stack is distinguished by the marker fill and line style: solid lines with fully filled markers denote the thin 0.5  $\mu\text{m}$  buffer, while dashed lines with hollow markers denote the thick 5.0  $\mu\text{m}$  buffer. Finally, the size of the scatter points scales with the lateral gate-to-drain spacing ( $L_{gd}$ ), progressing from small markers at 10  $\mu\text{m}$  to large markers at 20  $\mu\text{m}$ .

Trajectory lines connect device variants that share an identical vertical stack and material, isolating the effect of lateral geometric scaling. Directional arrows along these lines point from the shortest drift region to the longest, tracking the performance evolution as  $L_{gd}$  increases. As mapped by this logic, silicon configurations cluster strictly on the left side below the vertical dashed 1200 V target line, confirming their inability to meet the voltage requirement. In contrast, all compliant sapphire trajectories successfully cross the 1200 V boundary, scaling upward in breakdown voltage up to nearly 3000 V.

Crucially, the power plot reveals a fundamental design asymmetry: extending the lateral drift region ( $L_{gd}$ ) successfully increases breakdown voltage, but it incurs a severe penalty by shifting the device upward along the vertical axis, worsening the figure of merit product. Because on-state resistance scales linearly with  $L_{gd}$  while the output capacitance experiences only marginal reduction at high drain biases, acquiring additional voltage headroom beyond the qualification threshold via drift region extension yields diminishing returns. Consequently, the optimal design strategy identified by this power plot is to maintain the shortest possible drift length ( $L_{gd} = 10 \mu\text{m}$ ) to minimize conduction losses, while relying entirely on the native dielectric isolation of the thick sapphire substrate to achieve the required 1200 V blocking capability.

### 4.3 Field-Plate Architecture

Beyond geometry and substrate choice, shaping the surface electric field distribution is critical for high-voltage lateral transistors. In a lateral device without field plates, the electric field concentrates sharply at the drain-side edge of the gate, triggering premature breakdown. Field plates extend metal structures connected to the gate or source terminals to flatten this peak field distribution [8], [26]. This section presents the electrical performance of field-plated configurations on the sapphire platform. By evaluating variations from standard unplated devices to gate-connected and combined gate-and-source field-plate architectures, the following analysis characterizes the fundamental electrostatic trade-offs between electric field redistribution, breakdown voltage enhancement, and dynamic switching penalties. These structural enhancements complement the substrate optimization by managing surface electrostatics while the insulating sapphire substrate suppresses vertical leakage.

#### 4.3.1 Breakdown Voltage

A comprehensive comparison of the maximum breakdown voltage ( $V_{br}$ ) across the silicon baseline, standard sapphire architecture, and field-plated sapphire configurations as a function of gate-to-drain spacing ( $L_{gd}$ ) is compiled in Figure 4.20.

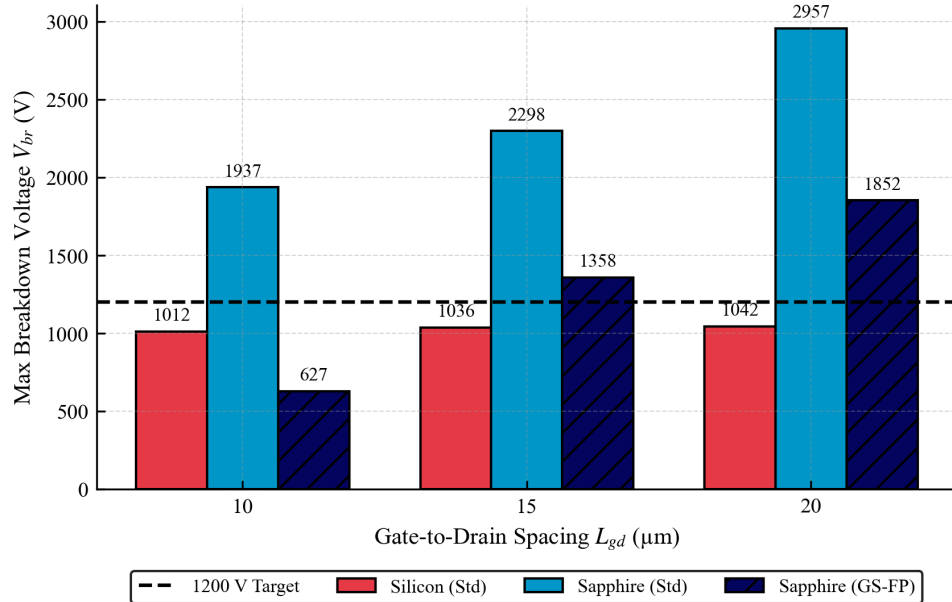


Figure 4.20: Breakdown voltage ( $V_{br}$ ) comparison across the silicon baseline, standard sapphire, and field-plated sapphire configurations.

The comparative data underscores the massive architectural advantage of eliminating the

conductive silicon backplane. While the silicon baseline plateaus near 1042 V due to premature vertical substrate leakage, the standard sapphire platform scales effectively, achieving breakdown voltages up to 2957 V at  $L_{gd} = 20\text{ }\mu\text{m}$ .

However, introducing field-plate architectures introduces an apparent anomaly wherein certain extracted  $V_{br}$  values exhibit a localized reduction compared to their unplated counterparts. Physically, field plates successfully mitigate surface electric field crowding at the gate edge, which should strictly enhance or maintain breakdown capability. This apparent downward trend in the extracted  $V_{br}$  data does not reflect a true physical limitation of the electrostatic shaping. Instead, it originates from severe numerical convergence challenges inherent to the finite element solver near the avalanche singularity. As the drain voltage ramps toward extreme levels in devices with complex multi-dielectric field plates, the localized steepness of the potential gradients combined with tight solver tolerances (METHOD block) can trigger premature matrix non-convergence. When the solver encounters severe grid-induced gradients that exceed local residual limits, the simulation terminates prematurely, capturing an artificial truncation current rather than the true physical onset of avalanche breakdown. Consequently, these specific data points represent a numerical solver limitation rather than an intrinsic device degradation mechanism.

### 4.3.2 Electrostatic Field Redistribution

To explicitly evaluate the efficacy of the top-side metallization, the internal electrostatic profile of the field-plated sapphire architecture is analyzed and directly compared against the unplated sapphire baseline. The primary function of a field plate is to act as a Faraday shield, redistributing the intense electric field in the access region to prevent premature breakdown and reliability degradation. By extending a conductive metal layer over the passivation, the internal equipotential lines are forced to spread laterally outward toward the drain.

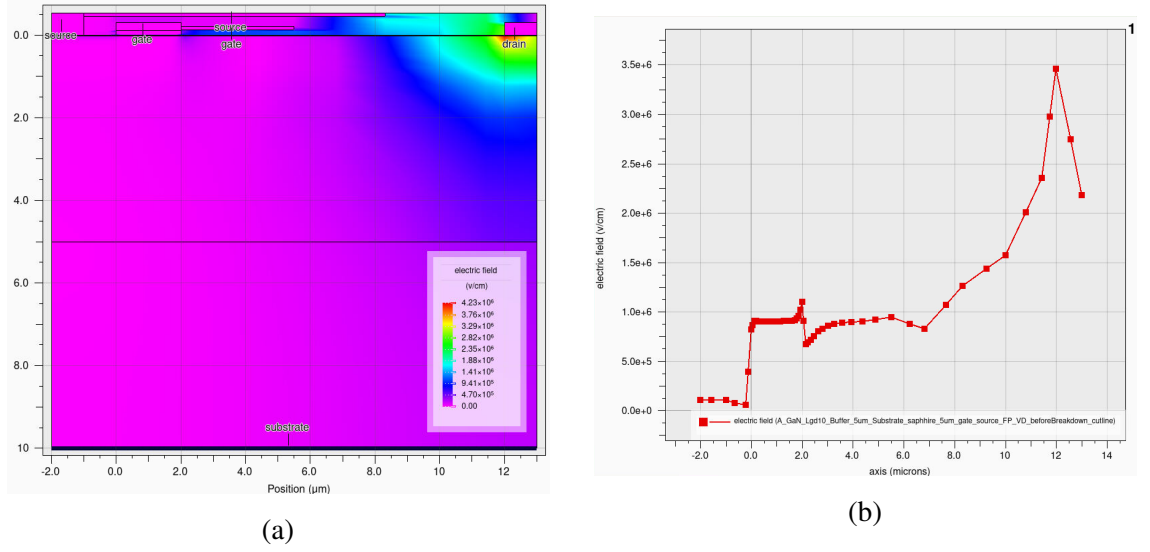


Figure 4.21: Electrostatic profile of the field-plated sapphire architecture prior to numerical solver termination, represented by (a) the two-dimensional electric field contour and (b) the one-dimensional electric field cutline. The top-side metallization successfully flattens the primary gate-edge field but introduces a secondary localized spike at the drain terminal.

The one-dimensional cutline confirms that the integrated gate and source field plates successfully achieve their fundamental design objective. In the standard unplated sapphire baseline, the electric field exhibited a catastrophic singularity exceeding  $5.0 \times 10^6$  V/cm directly at the gate edge. By contrast, the field-plate structure effectively suppresses this localized crowding. The electric field across the primary gate and secondary plate edges ( $x \approx 0$  to  $2 \mu\text{m}$ ) is remarkably well-distributed and flattened to a stable magnitude of approximately  $1.0 \times 10^6$  V/cm.

However, because the gate-edge field is so heavily suppressed, the lateral depletion region expands aggressively and without restriction across the entire drift region until it reaches the highly doped  $n^+$  drain contact. As visualized in the two-dimensional contour, the entire electrostatic burden is consequently shifted to the right side of the device. Since the depletion boundary is rigidly pinned against the highly conductive drain, any further applied voltage must be accommodated over a negligible spatial distance. This induces a massive localized field spike of approximately  $3.5 \times 10^6$  V/cm precisely at the drain terminal ( $x \approx 12 \mu\text{m}$ ). The field plates effectively protected the gate electrode but transferred the primary failure mechanism to the drain interface, a well-documented theoretical trade-off where the peak electric field shifts deeper into the drain access area as the plate length increases [8], [40].

Theoretically, successfully suppressing the primary gate-edge field crowding should yield a substantial improvement in the macroscopic avalanche breakdown voltage  $V_{br}$ . However, an unresolved anomaly emerges in the TCAD extraction for these plated structures.

The recorded  $V_{br}$  values for the field-plate variants routinely appear lower than their unplated counterparts. This apparent degradation does not reflect a true physical limitation of the electrostatic design. Instead, it is the result of a severe numerical convergence failure within the finite element solver.

The integration of complex multi-stepped field plates introduces sharp geometric corners at the metal-dielectric interfaces, which act as artificial crowding points where electric field contours reach non-physical peaks. Combined with the extreme localized gradients occurring at the drain terminal, these abrupt boundaries create severe mathematical stiffness in the impact ionization calculations. As the drain voltage scales and avalanche generation increases, the Jacobian matrix within the iterative Newton solver becomes ill-conditioned. The numerical residuals eventually exceed the strict solver error tolerances, causing the simulation to abort prematurely. Consequently, the recorded breakdown values represent the arbitrary voltage at which the mathematical solver failed to converge, fully masking the true physical avalanche enhancement provided by the field-plate architecture.

### 4.3.3 Dynamic Switching Characteristics on the Sapphire Platform

The dynamic performance of the GaN-on-sapphire HEMTs is evaluated by analyzing the highly non-linear, voltage-dependent terminal capacitances and their mathematical integrals. Because the sapphire substrate eliminates the conductive backplane present in the silicon baseline, the parasitic capacitance network is fundamentally altered. The integration of field-plate architectures introduces severe trade-offs between switching speed, stored energy, and gate-drive requirements.

### 4.3.4 Output Capacitance and Energy-Related Characteristics

The off-state output capacitance ( $C_{oss}$ ) determines the fundamental  $dv/dt$  limits during switching transitions. The integration of field plates introduces a severe capacitance penalty at low drain voltages. As visually evident in the  $C_{oss}$  profiles in Figure 4.22, the dashed curves representing field-plate architectures exhibit an order-of-magnitude higher capacitance near  $V_{ds} = 0$  V compared to the standard, unplated designs. This is physically caused by the large spatial overlap of the metal plate extensions over the dielectric passivation, which acts as a large parallel-plate capacitor coupled to the drain. However, as  $V_{ds}$  increases and the depletion region expands laterally past the field-plate edge,  $C_{oss}$  drops precipitously. At voltages exceeding 400 V, the field-plate variants actually demonstrate a lower absolute output capacitance than their unplated counterparts.

## 4 Results

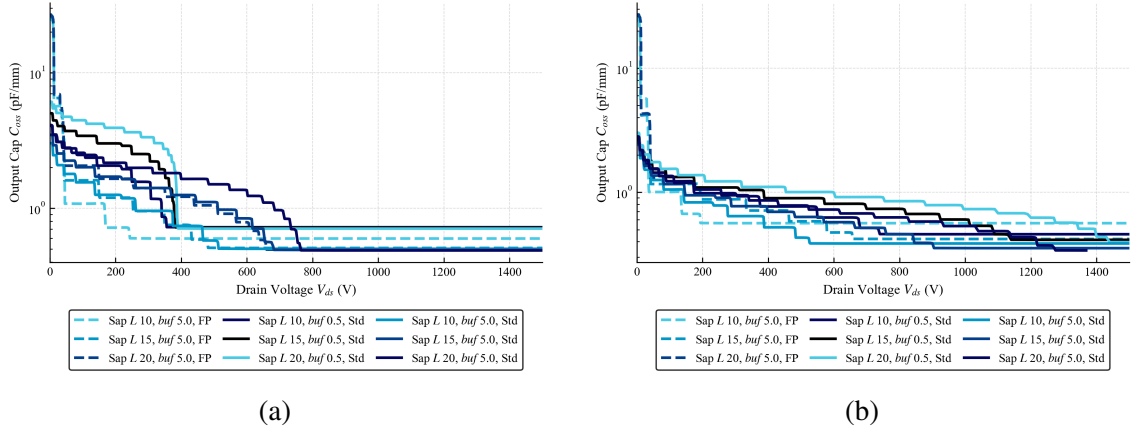


Figure 4.22: Non-linear output capacitance ( $C_{oss}$ ) profiles as a function of drain voltage ( $V_{ds}$ ) comparing field-plate integration on (a)  $5.0\ \mu\text{m}$  and (b)  $50.0\ \mu\text{m}$  sapphire substrates. Metal plate overlap severely inflates capacitance near zero bias, but rapid depletion yields lower absolute values above 400 V.

To standardize these highly non-linear profiles for circuit-level evaluation, the mathematically equivalent linear energy-related output capacitance ( $C_{o(er)}$ ) is extracted and plotted in Figure 4.23.

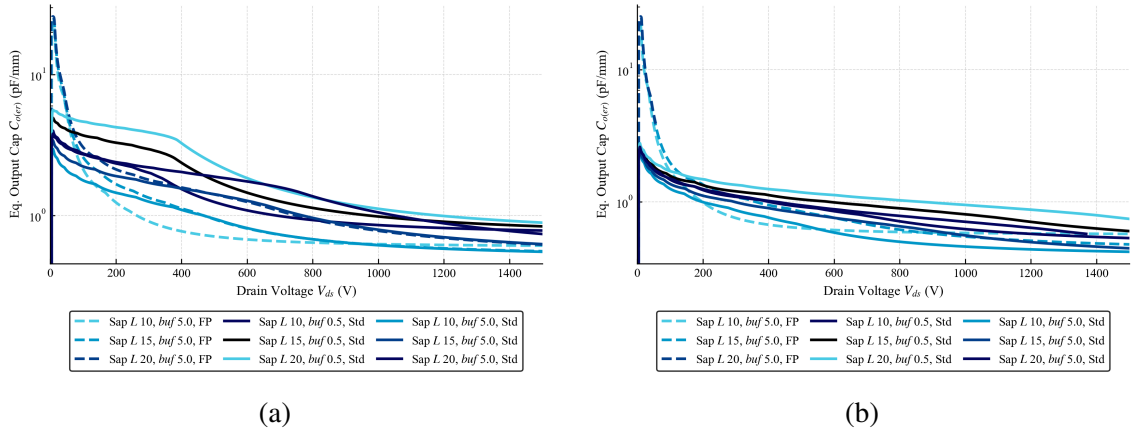


Figure 4.23: Energy-related output capacitance ( $C_{o(er)}$ ) profiles incorporating field-plate integration on (a)  $5.0\ \mu\text{m}$  and (b)  $50.0\ \mu\text{m}$  sapphire substrates.

This crossover behavior in the raw capacitance directly dictates the total large-signal switching losses, quantified by the output energy ( $E_{o(er)}$ ) in Figure 4.24. Because  $E_{o(er)}$  is calculated as the mathematical integral of  $V \cdot C_{oss}(V)$ , the capacitance at high voltages contributes exponentially more to the total stored energy. Consequently, despite the massive low-voltage capacitance penalty, the field-plate architectures yield a significantly lower total stored output energy at the 1200 V operating point. The rapid depletion of the plate capacitance results in a highly favorable energy profile, confirming that field-plate

integration on the sapphire platform improves avalanche breakdown without penalizing high-voltage turn-on losses.

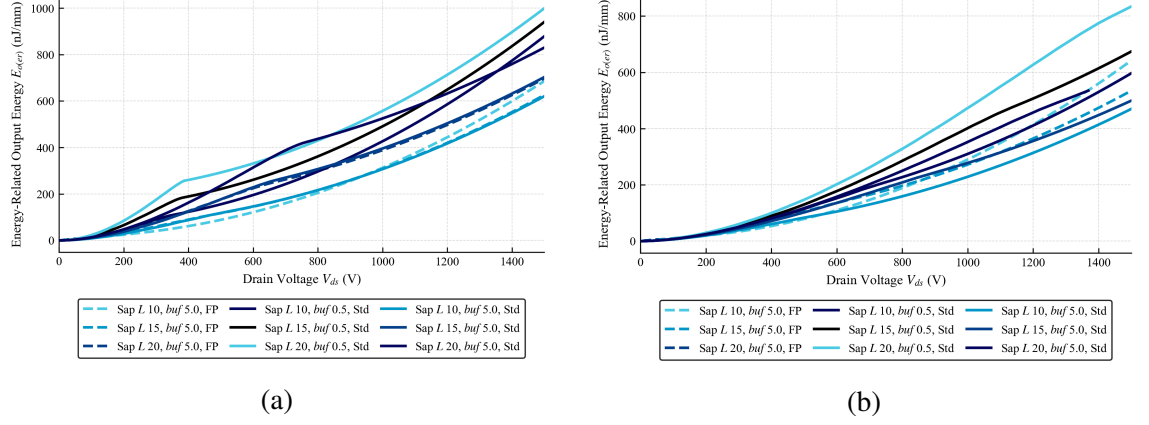


Figure 4.24: Total stored output energy ( $E_{o(er)}$ ) extracted from the voltage integral of  $C_{oss}$  for (a)  $5.0\ \mu\text{m}$  and (b)  $50.0\ \mu\text{m}$  sapphire substrates.

#### 4.3.5 Reverse Transfer Capacitance and Miller Charge

While output energy benefits from field-plate integration, the reverse transfer capacitance ( $C_{rss}$ ) suffers a compounding structural penalty. The gate-connected component of the field plate physically extends directly over the drain access region. This drastically elevates  $C_{rss}$  across the entire voltage sweep [25], [29], as plotted in Figure 4.25.

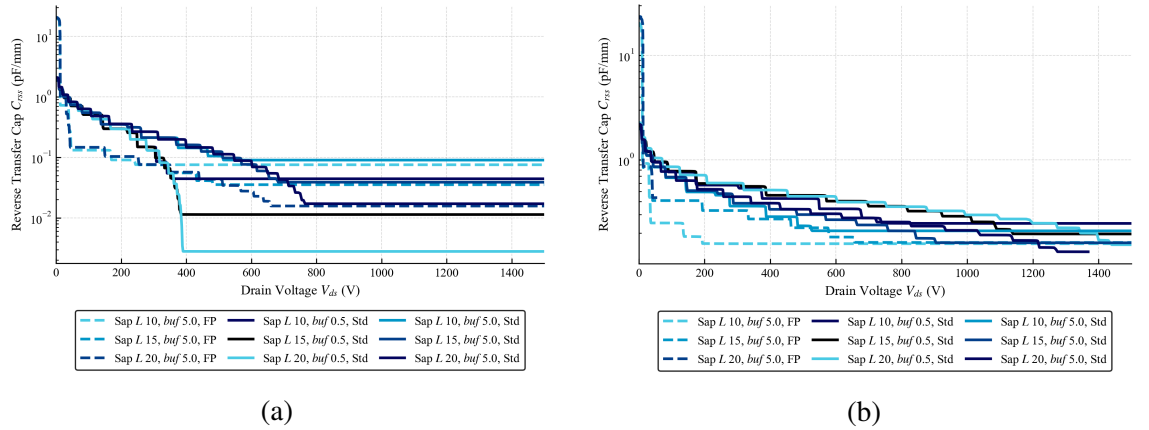


Figure 4.25: Reverse transfer capacitance ( $C_{rss}$ ) penalty induced by gate-connected field plates on (a)  $5.0\ \mu\text{m}$  and (b)  $50.0\ \mu\text{m}$  sapphire substrates. The spatial overlap of the gate extension strictly elevates feedback capacitance across the entire operational voltage range.

This sustained increase in  $C_{rss}$  maps directly to an inflated Miller charge ( $Q_{gd}$ ), illustrated in Figure 4.26. Unlike energy, charge is an unweighted integral of capacitance ( $Q = \int C dV$ ), meaning the massive low-voltage spike in  $C_{rss}$  is not mathematically diminished. The  $Q_{gd}$  required to switch the field-plate devices is substantially higher than the standard architectures under all geometric configurations. This elevated Miller charge prolongs the switching transient duration and increases gate-drive power requirements [25], representing the primary dynamic trade-off for the superior electrostatic field shaping provided by the metal plates.

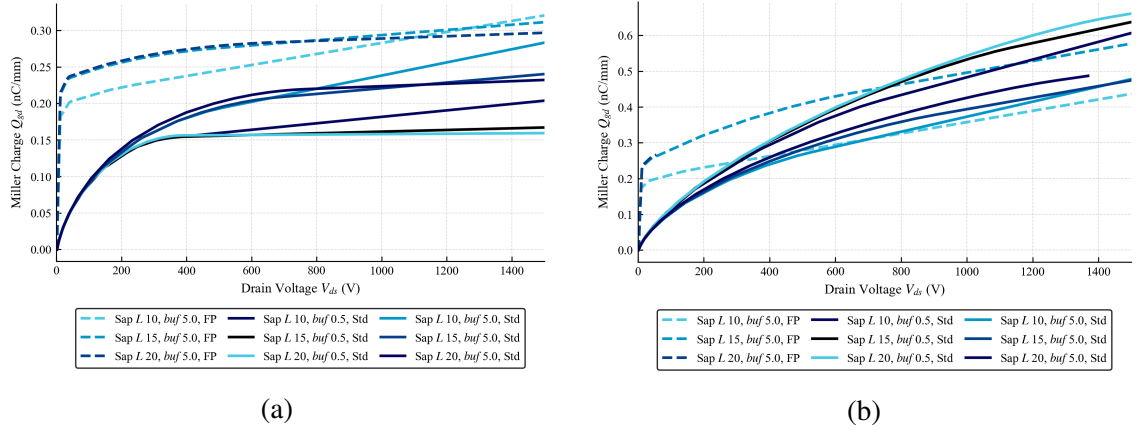


Figure 4.26: Total gate-to-drain Miller charge ( $Q_{gd}$ ) extracted from the unweighted integral of  $C_{rss}$  for (a) 5.0  $\mu\text{m}$  and (b) 50.0  $\mu\text{m}$  sapphire substrates.

## 5 Discussion

### 5.1 Silicon Substrate Performance and Limitations

The silicon baseline results settle the question of whether the 1200 V target is achievable on the incumbent substrate platform. It is not, and the reason is not a matter of insufficient drift region length or inadequate buffer thickness. It is a matter of device architecture.

The conductive silicon body beneath the GaN buffer does not behave as an inert mechanical support. It actively participates in the electrostatics of the device. Under high reverse bias, it acts as a fixed equipotential plane that compresses the entire voltage-blocking burden onto the GaN buffer layer alone. Once the vertical electric field across the buffer reaches the critical threshold, vertical current flows directly from the silicon body into the buffer via trap-assisted tunneling and Poole-Frenkel emission [13], [14], bypassing the lateral drift region entirely. The consequence is a breakdown voltage ceiling of 1043 V that is completely decoupled from lateral geometry. Extending  $L_{gd}$  from 10  $\mu\text{m}$  to 20  $\mu\text{m}$  yields a gain of only 30 V in breakdown voltage. Increasing the substrate thickness produces no change at all, since the conductive silicon body functions as an equipotential surface regardless of its depth and contributes nothing to vertical isolation. The same conductive backplane simultaneously inflates the output capacitance by collapsing the effective dielectric plate separation to the GaN buffer thickness alone. A thinner buffer reduces  $R_{ds,on}$  but collapses  $V_{br}$ ; a thicker buffer improves blocking but worsens capacitance. These two constraints pull in opposite directions and have no resolution within the silicon platform. The silicon results therefore do not point toward an optimized design. They point toward a platform change.

### 5.2 Sapphire Substrate as a Viable Alternative

Replacing silicon with an electrically insulating sapphire substrate removes the conductive backplane and fundamentally changes what the device can do. The substrate is no longer a fixed equipotential plane. It becomes an active dielectric layer that participates in voltage blocking alongside the GaN buffer. Equipotential contours that were compressed tightly into the buffer on silicon now spread smoothly through the full depth of the sapphire substrate, distributing the electrostatic burden over a much larger volume. This single material change simultaneously resolves both failure mechanisms identified in the silicon analysis.

To isolate the effect of substrate material alone, the two platforms are first compared at identical geometry:  $L_{gd} = 10\text{ }\mu\text{m}$  with a  $0.5\text{ }\mu\text{m}$  buffer. On silicon, the best achievable breakdown voltage at this geometry is  $754\text{ V}$  at  $31.2\text{ }\Omega\cdot\text{mm}$ . On sapphire at the same geometry, the breakdown voltage reaches  $1814\text{ V}$  at  $30.86\text{ }\Omega\cdot\text{mm}$ , a 141% improvement in blocking voltage at a conduction penalty of less than 1.1%, which is statistically indistinguishable from zero. The substrate material change alone, with no modification to the lateral geometry, epitaxial stack, or surface metallization, produces a device that clears the  $1200\text{ V}$  qualification threshold with a 51.2% margin, while the silicon equivalent falls 37.2% short. This comparison isolates the substrate variable without any confounding geometric factor and is the clearest demonstration that the silicon limitation is architectural rather than geometric.

A second comparison asks a different question: what is the best each platform can deliver under any combination of design variables available in this study? The best silicon configuration across the entire sweep ( $L_{gd} = 20\text{ }\mu\text{m}$ ,  $5.0\text{ }\mu\text{m}$  buffer) achieves  $V_{br} = 1043\text{ V}$  at  $R_{ds,on} = 89.7\text{ }\Omega\cdot\text{mm}$ . To reach even this maximum, the silicon platform requires doubling the drift region length and a ten-fold increase in buffer thickness relative to the optimal sapphire configuration. Despite these aggressive geometric concessions, the silicon platform still falls 13.1% short of the  $1200\text{ V}$  threshold and arrives there with an  $R_{ds,on}$  that is 190% higher than the best sapphire device. The best sapphire configuration, by contrast, achieves  $V_{br} = 1814\text{ V}$  at  $R_{ds,on} = 30.86\text{ }\Omega\cdot\text{mm}$ , representing 74% higher blocking voltage and 65.6% lower on-state resistance simultaneously, with a shorter drift region and thinner buffer. The sapphire platform therefore does not merely shift the performance frontier. It decouples blocking voltage from conduction loss in a way that silicon cannot replicate regardless of geometric optimisation, because the decoupling requires eliminating the conductive backplane rather than adjusting the device geometry above it.

The two comparisons together establish the argument on two levels. At identical geometry, sapphire delivers a 141% voltage gain at zero conduction penalty, proving the substrate is the governing variable. At best-case geometry for each platform, sapphire simultaneously improves both  $V_{br}$  and  $R_{ds,on}$  against a silicon configuration that has already exhausted every geometric degree of freedom available to it. In both cases the conclusion is the same: the  $1200\text{ V}$  target is structurally unreachable on silicon, and structurally straightforward on sapphire.

On the sapphire platform,  $L_{gd}$  extension is restored as a genuine voltage-scaling lever, with  $V_{br}$  scaling consistently from  $1937\text{ V}$  at  $L_{gd} = 10\text{ }\mu\text{m}$  to  $2633\text{ V}$  at  $L_{gd} = 20\text{ }\mu\text{m}$  at fixed buffer and substrate thickness. However,  $R_{ds,on}$  scales proportionally with drift length, worsening the figure of merit at longer geometries. The optimal strategy is therefore to maintain the shortest compliant drift region and rely on the native dielectric depth of the thick sapphire substrate for blocking voltage headroom, rather than extending  $L_{gd}$  and paying the conduction penalty.

### 5.3 Manufacturing and Economic Implications of Sapphire

The results carry implications beyond electrical figures of merit. The optimal configuration ( $L_{gd} = 10\text{ }\mu\text{m}$ ,  $0.5\text{ }\mu\text{m}$  buffer,  $500\text{ }\mu\text{m}$  sapphire substrate) is also the most economical to manufacture, and this is not a coincidence.

On silicon, suppressing vertical leakage requires a thick carbon-doped GaN buffer of at least  $4$  to  $5\text{ }\mu\text{m}$ , grown by metal-organic chemical vapour deposition under high thermal budgets with extended reactor times [15]. Removing this requirement on sapphire, where the insulating substrate itself provides vertical isolation without a thick compensating buffer, allows the buffer to be reduced to  $0.5\text{ }\mu\text{m}$ . This reduction directly cuts reactor run times, lowers precursor consumption, reduces parasitic chamber wall depositions, and decreases cleaning downtime between runs, all of which translate into lower cost per wafer in high-volume manufacturing. The  $500\text{ }\mu\text{m}$  sapphire substrate, which produces the highest breakdown voltages and the lowest output capacitance in the sweep, corresponds exactly to the standard as-grown thickness of commercially available sapphire wafers. No post-growth thinning, grinding, or chemical-mechanical polishing is required. The electrically optimal configuration and the most process-friendly configuration are therefore the same point in the design space, which is a strong practical argument in favour of the GaN-on-sapphire platform for  $1200\text{ V}$  power switching.

### 5.4 Field Plate Integration and Switching Performance

Eliminating vertical failure on the sapphire platform unmask the lateral gate-edge field crowding that was previously obscured on silicon. Without any electrostatic shaping, the one-dimensional electric field profile along the channel reveals a sharp peak exceeding  $5.0 \times 10^6\text{ V/cm}$  at the drain-side gate edge, a singularity that cannot be reduced by extending  $L_{gd}$  and that drives lateral breakdown irrespective of drift region length.

The dual gate-and-source-connected field plate directly targets this peak. By extending a grounded metal structure over the passivation dielectric into the drift region, the plate forces the equipotential lines to spread laterally beyond the gate edge, introducing a second transition zone deeper into the access region. The result, confirmed by the 1D cut-line, is a five-fold reduction in the gate-edge peak from above  $5.0 \times 10^6\text{ V/cm}$  to a stable plateau of approximately  $1.0 \times 10^6\text{ V/cm}$  across the gate and plate region. In a fabricated device, this redistribution is expected to yield higher physical breakdown voltage than the unplated baseline.

The switching analysis reveals that this electrostatic benefit is accompanied by a favourable dynamic trade-off. The field plate introduces a large capacitance spike at low drain voltages due to the metal plate overlap over the passivation, but this capacitance

depletes rapidly as the drain voltage rises. Above 400 V, the field-plated device exhibits lower  $C_{o(er)}$  and lower stored output energy  $E_{o(er)}$  than the unplated baseline. Since hard-switching energy loss is dominated by the capacitance at high voltages, the net effect is a reduction in switching losses at the operating point despite the low-voltage capacitance penalty. The cost of this improvement is an increase in reverse transfer capacitance  $C_{rss}$  and gate-to-drain charge  $Q_{gd}$ , which elevates gate-drive requirements and prolongs the Miller plateau. This is the genuine trade-off of the field plate approach and represents a circuit-level design constraint rather than a device-level failure.

The numerically extracted  $V_{br}$  values for the field-plated configurations appear lower than the unplated baseline due to finite-element solver convergence failure near the avalanche singularity. The multi-dielectric geometry of the field plate introduces sharp geometric corners that create mathematical stiffness in the impact ionisation calculation, causing the solver to abort prematurely before true physical breakdown is reached. The 1D electric field evidence confirms that the gate-edge field is well below the critical threshold at equivalent bias conditions, and that the physical device would sustain higher breakdown voltage with the field plate in place than without it.

## 6 Conclusion

This thesis investigated the feasibility of extending lateral GaN HEMT technology to the 1200 V power switching class through a systematic TCAD parametric sweep across substrate material, gate-to-drain spacing, buffer thickness, substrate thickness, and field-plate architecture. The study was motivated by the growing demand for high-voltage power semiconductors in electric vehicle and renewable energy applications, and by the fundamental inability of the GaN-on-silicon platform to meet this demand despite the material's intrinsic capability.

The silicon baseline established that vertical punch-through imposes a hard breakdown voltage ceiling of 1043 V that is insensitive to all geometric adjustments within the platform. The most aggressive silicon configuration available in this study ( $L_{gd} = 20\text{ }\mu\text{m}$ ,  $5.0\text{ }\mu\text{m}$  buffer) reaches this maximum at the cost of  $R_{ds,on} = 89.7\text{ }\Omega\cdot\text{mm}$  and still falls 13.1% short of the 1200 V threshold. Extending  $L_{gd}$  from  $10\text{ }\mu\text{m}$  to  $20\text{ }\mu\text{m}$  yields a gain of only 30 V in breakdown voltage, and substrate thickness variation produces no change at all. Silicon does not point toward an optimized design. It points toward a platform change.

Substituting an electrically insulating sapphire substrate resolved both failure mechanisms simultaneously. At identical geometry ( $L_{gd} = 10\text{ }\mu\text{m}$ ,  $0.5\text{ }\mu\text{m}$  buffer), sapphire achieves  $V_{br} = 1814\text{ V}$  against silicon's 754 V, a 141% improvement, while  $R_{ds,on}$  remains virtually unchanged at  $30.86\text{ }\Omega\cdot\text{mm}$  versus  $31.2\text{ }\Omega\cdot\text{mm}$ . Comparing the best device of each platform irrespective of geometry reinforces this finding: the best sapphire configuration simultaneously delivers 74% higher blocking voltage and 65.6% lower on-state resistance than the best silicon configuration, which despite using a longer drift region and thicker buffer still cannot qualify. Fifteen of eighteen sapphire configurations exceed the 1200 V threshold,  $C_{o(er)}$  is reduced by up to 31.9% at the 1200 V evaluation bias, and the optimal design point ( $L_{gd} = 10\text{ }\mu\text{m}$ ,  $0.5\text{ }\mu\text{m}$  buffer,  $500\text{ }\mu\text{m}$  substrate) achieves a 51.2% qualification margin while simultaneously offering the lowest manufacturing cost in the sweep through a thin epitaxial buffer and a substrate thickness that matches the standard as-grown commercial sapphire wafer dimension.

The integration of a dual gate-and-source-connected field plate on the sapphire platform suppressed the gate-edge electric field peak from above  $5.0 \times 10^6\text{ V/cm}$  to approximately  $1.0 \times 10^6\text{ V/cm}$ , a five-fold reduction confirming that the field-plate architecture achieves its primary electrostatic design objective and is expected to yield higher physical breakdown voltage in a fabricated device. Field-plate integration simultaneously reduced  $C_{o(er)}$  and stored output energy  $E_{o(er)}$  above 400 V, improving the hard-switching figure of merit.

The trade-off is an increase in  $Q_{gd}$  that elevates gate-drive requirements, which represents a circuit-level design constraint rather than a device-level failure.

The three-tier performance hierarchy established by this work provides a coherent design roadmap for 1200 V lateral GaN HEMTs. Silicon fails unconditionally due to a structural architectural limitation. Sapphire meets and exceeds the voltage target with a simultaneous improvement in switching efficiency and a manufacturing cost advantage. The dual field plate further improves lateral field management and reduces switching energy losses on the sapphire platform. The results identify the GaN-on-sapphire platform with dual field plate metallization as a technically viable and economically competitive candidate for experimental fabrication targeting this voltage class.

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